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Methods and instrumentation for intervertebral cage expansion

Abstract

An instrument may be coupled to a multi-axis expandable intervertebral cage so that the cage may be inserted into an intervertebral space, expanded along multiple different directions, filled with bone graft, and locked with a fastener. The instrument may be part of an instrument set that includes auxiliary instruments to determine implant size, insert bone graft into the cage, and deliver the fastener.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 15/793,950 filed on Oct. 25, 2017, entitled “Methods and Instrumentation for Intervertebral Cage Expansion”, which claims the benefit of U. S. Provisional Patent Application No. 62/412,781, filed on Oct. 25, 2016, entitled “Methods and Instrumentation for Intervertebral Cage Expansion”. The foregoing is incorporated by reference as though set forth herein in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to instruments and methods for implantation and expansion of an intervertebral implant. The instruments and methods described herein may be used to expand an intervertebral implant or spacer which is capable of being expanded in one or more directions: horizontally (laterally) and/or vertically (cephalocaudally). More specifically, an embodiment of the instruments and methods described herein may be used to implant and expand the intervertebral cages described in pending U. S. patent application Ser. No. 15/244,446, entitled EXPANDABLE INTERVERTEBRAL IMPLANTS, filed on Aug. 23, 2016, which is incorporated by reference as though set forth in its entirety.

BACKGROUND

(2) There is a need for an instrument that may be coupled to a multi-axis expandable intervertebral cage so that the cage may be inserted into an intervertebral space, expanded along multiple different directions, filled with bone graft, and/or locked with a fastener. There is also a need for an

instrument set that includes the implant insertion instrument and auxiliary instruments to determine implant size, insert bone graft into the cage, and deliver the fastener.

(3) The implant insertion instruments disclosed herein address certain characteristics of the expandable intervertebral cages. The cages expand horizontally and/or vertically as first and second implant end bodies are drawn together along a longitudinal instrument axis. Each of the implant insertion instruments includes a connector that couples to the first implant end body and a draw bar that couples to the second implant end body. The draw bar may be rotatable about the instrument axis to connect with and disconnect from the second implant end body, however the draw bar does not rotate about the axis during implant expansion. In some examples, in order to expand the implant, the implant insertion instrument may first be actuated to lock the draw bar to prevent axial rotation, then actuated to accomplish implant expansion. These actuations may be coupled together so as to occur together.

(4) In some examples, the implant insertion instruments disclosed herein also address certain characteristics of specific approaches to the intervertebral space in the spine. For example, in transforaminal lumbar interbody fusion (TLIF), a patient may be in the prone position (face down) and the approach trajectory may lie in the transverse plane of the operative intervertebral space and inclined about 30 degrees from the sagittal plane (vertical) or 60 degrees from the coronal plane (horizontal). The weight of the implant insertion instrument thus causes a nontrivial force moment on the implant which tends to rotate the implant laterally out of position. Disclosed implant insertion instruments have removable handle assemblies to reduce instrument weight and thus the force moment on the implant.

(5) In some examples, the implant insertion instruments disclosed herein also provide a useful and convenient collateral function for a graft funnel, wherein the funnel may be used to unthread an implant retainer shaft after bone graft has been delivered to the operative site.

SUMMARY

(6) The various systems and methods of the present technology have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available expandable implants and related instruments. The systems and methods of the present technology may provide improved instruments and methods for implanting expandable implants, such as spinal interbody fusion spacers.

(7) To achieve the foregoing, and in accordance with the technology as embodied and broadly described herein, in an aspect of the technology, an instrument includes: a first instrument part that is removably connectable to a first implant part of an implant; and a second instrument part that is removably connectable to a second implant part of the implant; wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along a central longitudinal first axis of the instrument without rotating about the first axis during instrument actuation for implant expansion.

(8) Embodiments of this aspect may include any or all of the following attributes. The first instrument part includes a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the first axis; wherein the instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the first axis; wherein when the rotation lock is engaged, the draw bar cannot rotate about the first axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented

from rotating about the first axis. The rotation lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is engaged and the actuator disengages from the draw bar when the rotation lock is disengaged. The draw bar includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the draw bar, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the first axis. The first instrument part includes a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the first axis; wherein the instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the draw bar are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis; The instrument of claim 5, wherein the shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis.

(9) In another aspect of the technology, an instrument includes a first instrument part that is removably connectable to a first implant part of an implant by rotation of the first instrument part relative to the first implant part about a central longitudinal axis of the instrument; and a second instrument part that is removably connectable to a second implant part of the implant; wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along the central longitudinal axis of the instrument so that the first implant part approaches the second implant part, wherein the first instrument part is prevented from rotating about the central longitudinal axis while the first instrument part and the first implant part are translating.

(10) Embodiments of this aspect may include any or all of the following attributes. The instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the first instrument part to translate the first instrument part relative to the second instrument part and the second implant part along the first axis; wherein when the rotation lock is engaged, the first instrument part is prevented from rotating about the central longitudinal axis, wherein when the rotation lock is disengaged, the first instrument part is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the first instrument part, the rotation lock is disengaged, and the first instrument part is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the first instrument part, the rotation lock is engaged, and the first instrument part is prevented from rotating about the first axis. The rotation

lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the first instrument part are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the first instrument part when the rotation lock is engaged and the actuator disengages from the first instrument part when the rotation lock is disengaged. The first instrument part includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the first instrument part, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the first instrument part threads to translate the first instrument part relative to the second instrument part and the second implant part along the first axis. The instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the first instrument part are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis; The shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the first instrument part relative to the second instrument part and the second implant part along the first axis.

(11) In yet another aspect of the technology, an instrument includes a draw bar that is removably connectable to a first implant part of an implant by rotation of the draw bar relative to the first implant part about a central longitudinal axis of the instrument, wherein the draw bar is carried at a first end of a first shaft; and a second instrument part that is removably connectable to a second implant part of the implant, wherein the second instrument part is carried at a first end of a second shaft; wherein the instrument is actuatable to translate the draw bar toward the second instrument part along the central longitudinal axis so that the draw bar does not rotate about the central longitudinal axis during instrument actuation.

(12) Embodiments of this aspect may include any or all of the following attributes. The instrument further includes an actuator and a rotation lock; wherein instrument actuation includes movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the first axis; wherein when the rotation lock is engaged, the draw bar cannot rotate about the first axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the first axis; wherein the instrument includes a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the first axis, wherein the instrument includes a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented from rotating about the first axis. The rotation lock includes a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the first axis, wherein the spline insert is fixed in translation along and rotation about the first axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the

rotation lock ring cannot rotate about the first axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that the rotation lock ring is free to rotate about the first axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is engaged and the actuator disengages from the draw bar when the rotation lock is disengaged. The draw bar includes threads; wherein the actuator includes a threaded slider, wherein the threaded slider includes threads that engage the threads of the draw bar, wherein instrument actuation includes rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the first axis. The instrument further includes a shuttle and an actuator; wherein the shuttle is fixed in rotation about the first axis, wherein the shuttle and the draw bar are fixed together for translation along the first axis, wherein instrument actuation includes movement of the actuator in engagement with the shuttle to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes threads, wherein the actuator includes a threaded driver, wherein the threaded driver includes threads that engage the threads of the shuttle, wherein instrument actuation includes rotation of the threaded driver threads in engagement with the shuttle threads to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis. The shuttle includes a series of transverse grooves, wherein the actuator includes a ratchet mechanism, wherein the ratchet mechanism includes a drive link that engages the series of transverse grooves, wherein instrument actuation includes translation of the drive link in engagement with the series of transverse grooves to translate the shuttle and the draw bar relative to the second instrument part and the second implant part along the first axis.

(13) These and other features and advantages of the present technology will become more fully apparent from the following description and appended claims, or may be learned by the practice of the technology as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the technology will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the technology, the exemplary embodiments will be described with additional specificity and detail through use of the accompanying drawings in which:

(2) FIG. 1 is a top view of an insertion and expansion instrument;

(3) FIG. 2 is a side view of the instrument of FIG. 1 with a draw bar assembly inserted into the instrument and an expandable intervertebral implant connected to a distal end of the instrument;

(4) FIG. 3 is a cross-sectional view of the instrument and implant of FIG. 2 taken along line 3-3 of FIG. 2;

(5) FIG. 4 is a distal detail cross-sectional view of the instrument and implant of FIG. 3;

(6) FIG. 5 is a proximal detail cross-sectional view of the instrument of FIG. 3;

(7) FIG. 6 is a top view of the instrument of FIG. 1 in a first partially assembled state;

(8) FIG. 7 is a top view of the instrument of FIG. 1 in a second partially assembled state;

(9) FIG. 8 is a top view of the instrument of FIG. 1 in a fully assembled state;

(10) FIG. 9 is an exploded perspective view of the instrument of FIG. 1;

(11) FIG. 10 is another exploded perspective view of the instrument of FIG. 1, from a different direction;

(12) FIG. 11 is an exploded perspective view of a shuttle assembly of the instrument of FIG. 1;

- (13) FIG. 12 is another exploded perspective view of the shuttle assembly of FIG. 11, from a different direction;
- (14) FIG. 13 is a proximal end view of an inserter handle of the instrument of FIG. 9;
- (15) FIG. 14 is a distal end view of the inserter handle of the instrument of FIG. 9;
- (16) FIG. 15 is a proximal end view of a selector ring of the instrument of FIG. 9;
- (17) FIG. 16 is a proximal end view of an expansion knob of the instrument of FIG. 9;
- (18) FIG. 17 is a distal end view of the expansion knob of the instrument of FIG. 9;
- (19) FIG. 18 is a proximal end view of a threaded driver of the instrument of FIG. 9;
- (20) FIG. 19 is a distal end view of the threaded driver of the instrument of FIG. 9;
- (21) FIG. 20 is a proximal end view of a lock tube coupling of the lock tube assembly of FIG. 24;
- (22) FIG. 21 is a distal end view of the lock tube coupling of the lock tube assembly of FIG. 24;
- (23) FIG. 22 is an exploded perspective view of a shaft assembly of the instrument of FIG. 1;
- (24) FIG. 23 is another exploded perspective view of the shaft assembly of FIG. 22, from a different direction;
- (25) FIG. 24 is an exploded perspective view of a lock tube assembly of the instrument of FIG. 1;
- (26) FIG. 25 is another exploded perspective view of the lock tube assembly of FIG. 24, from a different direction;
- (27) FIG. 26 is a top view of the draw bar assembly of FIG. 2;
- (28) FIG. 27 is a top view of a tamp assembly;
- (29) FIG. 28 is a top view of a funnel;
- (30) FIG. 29 is a top view of a screwdriver assembly coupled to a lockout screw;
- (31) FIG. 30 is a distal detail view of the screwdriver assembly and lockout screw of FIG. 29;
- (32) FIG. 31 is an exploded perspective view of the draw bar assembly of FIG. 26;
- (33) FIG. 32 is another exploded perspective view of the draw bar assembly of FIG. 26, from a different direction;
- (34) FIG. 33 is a perspective view of the funnel of FIG. 28;
- (35) FIG. 34 is another perspective view of the funnel of FIG. 28, from a different direction;
- (36) FIG. 35 is an exploded perspective view of the tamp assembly of FIG. 27;
- (37) FIG. 36 is another exploded perspective view of the tamp assembly of FIG. 27, from a different direction;
- (38) FIG. 37 is an exploded perspective view of the screwdriver assembly of FIG. 29;
- (39) FIG. 38 is another exploded perspective view of the screwdriver assembly of FIG. 29, from a different direction;
- (40) FIG. 39 is a perspective view of a screw holder and a screw holder catch of the screwdriver assembly of FIG. 29;
- (41) FIG. 40 is another perspective view of a screw holder and a screw holder catch of the screwdriver assembly of FIG. 29, from a different direction;
- (42) FIG. 41 is a proximal end view of the screw holder of FIG. 40;
- (43) FIG. 42 is a distal end view of the screw holder of FIG. 40;
- (44) FIG. 43 is a perspective view of the lockout screw of FIG. 29;
- (45) FIG. 44 is another perspective view of the lockout screw of FIG. 29, from a different direction;
- (46) FIG. 45 is a rear or second end view of the implant of FIG. 2 in a collapsed configuration;
- (47) FIG. 46 is a top view of the implant of FIG. 2;
- (48) FIG. 47 is a side view of the implant of FIG. 2;
- (49) FIG. 48 is an isometric view of the instrument of FIG. 1 in an unlocked configuration;
- (50) FIG. 49 is a closeup view of a distal end of the instrument of FIG. 48;
- (51) FIG. 50 is an isometric view of the instrument of FIG. 1 with the implant of FIG. 2 connected to the distal end of the instrument;
- (52) FIG. 51 is a closeup view of the distal end of the instrument and implant of FIG. 50;
- (53) FIG. 52 is an isometric view of the instrument and implant of FIG. 50, the implant adjacent a

vertebral body and in a laterally expanded configuration;

(54) FIG. 53 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 52;

(55) FIG. 54 is an isometric view of the instrument and implant of FIG. 52, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration;

(56) FIG. 55 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 54;

(57) FIG. 56 is an isometric view of the instrument and implant of FIG. 54, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration, and the funnel of FIG. 28 connected to a proximal end of the instrument;

(58) FIG. 57 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 56;

(59) FIG. 58 is an isometric view of the instrument and implant of FIG. 56, the implant adjacent a vertebral end body and in a laterally and vertically expanded configuration, the lockout screw of FIG. 29 extending through the implant and the screwdriver assembly of FIG. 29 inserted through the instrument;

(60) FIG. 59 is a closeup view of the distal end of the instrument, implant, and vertebra of FIG. 58;

(61) FIG. 60 is a top view of the implant and vertebra of FIG. 58;

(62) FIG. 61 is an anterior view of the implant and vertebra of FIG. 58;

(63) FIG. 62 is a right lateral view of the implant and vertebra of FIG. 58;

(64) FIG. 63 is a left lateral view of the implant and vertebra of FIG. 58;

(65) FIG. 64 is a top view of another insertion and expansion instrument;

(66) FIG. 65 is a cross-sectional view of the instrument of FIG. 64, taken along line 65-65 of FIG. 64;

(67) FIG. 66 is a proximal detail cross-sectional view of the instrument of FIG. 65;

(68) FIG. 67 is a distal detail cross-sectional view of the instrument of FIG. 65;

(69) FIG. 68 is a perspective view of the instrument of FIG. 64;

(70) FIG. 69 is another perspective view of the instrument of FIG. 64, from a different direction;

(71) FIG. 70 is an exploded perspective view of the instrument of FIG. 64;

(72) FIG. 71 is another exploded perspective view of the instrument of FIG. 64, from a different direction;

(73) FIG. 72 is an exploded perspective view of a ratchet mechanism of the instrument of FIG. 64;

(74) FIG. 73 is another exploded perspective view of the ratchet mechanism of FIG. 72, from a different direction;

(75) FIG. 74 is an exploded perspective view of a shaft assembly of the instrument of FIG. 64;

(76) FIG. 75 is another exploded perspective view of the shaft assembly of FIG. 74, from a different direction;

(77) FIG. 76 is an exploded perspective view of a lock tube assembly of the instrument of FIG. 64;

(78) FIG. 77 is another exploded perspective view of the lock tube assembly of FIG. 76, from a different direction;

(79) FIG. 78 is an exploded perspective view of a draw bar assembly of the instrument of FIG. 64;

(80) FIG. 79 is another exploded perspective view of the draw bar assembly of FIG. 78, from a different direction;

(81) FIG. 80 is an exploded perspective view of a handle, a lock lever, a stop screw, and a release ring of the instrument of FIG. 64;

(82) FIG. 81 is another exploded perspective view of the handle, lock lever, stop screw, and release ring of FIG. 80, from a different direction;

(83) FIG. 82 is a proximal end view of an inserter handle of the instrument of FIG. 70;

(84) FIG. 83 is a distal end view of the inserter handle of the instrument of FIG. 70;

(85) FIG. 84 is a right side view of the inserter handle of the instrument of FIG. 70;

(86) FIG. 85 is a left side view of the inserter handle of the instrument of FIG. 70;

(87) FIG. 86 is a top view of another insertion and expansion instrument coupled to another expandable intervertebral implant;

(88) FIG. **87** is a side view of the instrument and implant of FIG. **86**;

(89) FIG. **88** is a cross-sectional view of the instrument and implant of FIG. **87**, taken along line **88-88** of FIG. **87**;

(90) FIG. **89** is a cross-sectional view of the instrument and implant of FIG. **86**, taken along line **89-89** of FIG. **86**;

(91) FIG. **90** is a perspective view of the instrument of FIG. **86**;

(92) FIG. **91** is another perspective view of the instrument of FIG. **86**, from a different direction;

(93) FIG. **92** is a perspective view of an inserter connector of the instrument of FIG. **86**;

(94) FIG. **93** is another perspective view of the inserter connector of FIG. **92**, from a different direction;

(95) FIG. **94** is a perspective view of a connector plate of the instrument of FIG. **86**;

(96) FIG. **95** is another perspective view of the connector plate of FIG. **94**, from a different direction;

(97) FIG. **96** is a perspective view of an inserter shaft of the instrument of FIG. **86**;

(98) FIG. **97** is another perspective view of the inserter shaft of FIG. **96**, from a different direction;

(99) FIG. **98** is a perspective view of an implant retainer shaft of the instrument of FIG. **86**;

(100) FIG. **99** is another perspective view of the implant retainer shaft of FIG. **98**, from a different direction;

(101) FIG. **100** is a perspective view of connector buttons of the instrument of FIG. **86**;

(102) FIG. **101** is another perspective view of the connector buttons of FIG. **100**, from a different direction;

(103) FIG. **102** is a perspective view of a thumb wheel of the instrument of FIG. **86**;

(104) FIG. **103** is another perspective view of the thumb wheel of FIG. **102**, from a different direction;

(105) FIG. **104** is a perspective view of a wheel axle of the instrument of FIG. **86**;

(106) FIG. **105** is another perspective view of the wheel axle of FIG. **104**, from a different direction;

(107) FIG. **106** is a perspective view of an inserter handle of the instrument of FIG. **86**;

(108) FIG. **107** is another perspective view of the inserter handle of FIG. **106**, from a different direction;

(109) FIG. **108** is a proximal end view of the inserter handle of FIG. **106**;

(110) FIG. **109** is a perspective view of a strike plate of the instrument of FIG. **86**;

(111) FIG. **110** is another perspective view of the strike plate of FIG. **109**, from a different direction;

(112) FIG. **111** is a perspective view of a spline insert of the instrument of FIG. **86**;

(113) FIG. **112** is another perspective view of the spline insert of FIG. **111**, from a different direction;

(114) FIG. **113** is a perspective view of a rotation lock ring of the instrument of FIG. **86**;

(115) FIG. **114** is another perspective view of the rotation lock ring of FIG. **113**, from a different direction;

(116) FIG. **115** is a perspective view of a cam lever of the instrument of FIG. **86**;

(117) FIG. **116** is another perspective view of the cam lever of FIG. **115**, from a different direction;

(118) FIG. **117** is a perspective view of a slider button of the instrument of FIG. **86**;

(119) FIG. **118** is another perspective view of the slider button of FIG. **117**, from a different direction;

(120) FIG. **119** is a side view of the slider button of FIG. **117**;

(121) FIG. **120** is a perspective view of a threaded slider of the instrument of FIG. **86**;

(122) FIG. **121** is another perspective view of the threaded slider of FIG. **120**, from a different direction;

(123) FIG. **122** is a perspective view of a knob top part of the instrument of FIG. **86**;

(124) FIG. **123** is another perspective view of the knob top part of FIG. **122**, from a different direction;

(125) FIG. **124** is a perspective view of a knob bottom part of the instrument of FIG. **86**;

(126) FIG. **125** is another perspective view of the knob bottom part of FIG. **124**, from a different direction;

(127) FIG. **126** is a perspective view of a draw bar assembly for use with the instrument of FIG. **86**;

(128) FIG. **127** is another perspective view of the draw bar assembly of FIG. **126**, from a different direction;

(129) FIG. **128** is an exploded perspective view of the draw bar assembly of FIG. **126**;

(130) FIG. **129** is another exploded perspective view of the draw bar assembly of FIG. **126**, from a different direction;

(131) FIG. **130** is a perspective view of a trial assembly for use with the instrument of FIG. **86**;

(132) FIG. **131** is another perspective view of the trial assembly of FIG. **130**, from a different direction;

(133) FIG. **132** is an exploded perspective view of the trial assembly of FIG. **130**;

(134) FIG. **133** is another exploded perspective view of the trial assembly of FIG. **130**, from a different direction;

(135) FIG. **134** is a perspective view of a funnel for use with the instrument of FIG. **86**;

(136) FIG. **135** is another perspective view of the funnel of FIG. **134**, from a different direction;

(137) FIG. **136** is a perspective view of a screwdriver assembly for use with the instrument of FIG. **86**;

(138) FIG. **137** is another perspective view of the screwdriver assembly of FIG. **136**, from a different direction;

(139) FIG. **138** is an exploded perspective view of the screwdriver assembly of FIG. **136**;

(140) FIG. **139** is another exploded perspective view of the screwdriver assembly of FIG. **136**, from a different direction;

(141) FIG. **140** is a rear or second end view of the implant of FIG. **86** in a collapsed configuration;

(142) FIG. **141** is a top view of the implant of FIG. **86**;

(143) FIG. **142** is a side view of the implant of FIG. **86**;

(144) FIG. **143** is a perspective view of the trial assembly of FIG. **130** adjacent a vertebral body;

(145) FIG. **144** is a distal detail perspective view of the trial assembly and vertebral body of FIG. **143**;

(146) FIG. **145** is a perspective view of the instrument and implant of FIG. **86** coupled to the draw bar assembly of FIG. **126**, the slider button of the instrument in a proximal position;

(147) FIG. **146** is a distal detail perspective view of the instrument and implant of FIG. **145**;

(148) FIG. **147** is a perspective view of the instrument, draw bar assembly, and implant of FIG. **145**, the slider button of the instrument in a distal position;

(149) FIG. **148** is a distal detail perspective view of the instrument and implant of FIG. **147**;

(150) FIG. **149** is a perspective view of the instrument, draw bar assembly, and implant of FIG. **144**, the implant adjacent a vertebral body and in a laterally expanded configuration;

(151) FIG. **150** is a distal detail perspective view of the instrument, implant, and vertebra of FIG. **149**;

(152) FIG. **151** is a perspective view of the instrument, draw bar assembly, and implant of FIG. **149**, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration;

(153) FIG. **152** is a distal detail perspective view of the instrument, implant, and vertebra of FIG. **151**;

(154) FIG. **153** is a perspective view of the instrument, draw bar assembly, and implant of FIG. **151**, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, the slider button of the instrument released;

(155) FIG. **154** is a distal detail perspective view of the instrument, implant, and vertebra of FIG.

153;

(156) FIG. 155 is a perspective view of the instrument, draw bar assembly, and implant of FIG. 153, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, the slider button in the proximal position;

(157) FIG. 156 is a distal detail perspective view of the instrument, implant, and vertebra of FIG. 155;

(158) FIG. 157 is a perspective view of the instrument and implant of FIG. 155, the implant in a laterally and vertically expanded configuration, a handle assembly of the instrument partially removed from a shaft assembly of the instrument;

(159) FIG. 158 is a perspective view of the shaft assembly and implant of FIG. 157, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, with the funnel of FIG. 134 and the tamp assembly of FIG. 27 coupled to the shaft assembly;

(160) FIG. 159 is a perspective view of the shaft assembly, implant, and funnel of FIG. 158, the implant adjacent a vertebral body and in a laterally and vertically expanded configuration, with the screwdriver assembly of FIG. 136 coupled to the lockout screw of FIG. 43 and oriented for insertion into the shaft assembly;

(161) FIG. 160 is a perspective view of the implant and lockout screw of FIG. 159 adjacent a vertebral body and in a laterally and vertically expanded configuration;

DETAILED DESCRIPTION

(162) Exemplary embodiments of the technology will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the technology, as generally described and illustrated in the figures herein, could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the apparatus, system, and method is not intended to limit the scope of the invention, as claimed, but is merely representative of exemplary embodiments of the technology.

(163) The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “abutting” refers to items that are in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two features that are connected such that a fluid within one feature is able to pass into the other feature.

(164) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

(165) Standard medical planes of reference and descriptive terminology are employed in this specification. A sagittal plane divides a body into right and left portions. A mid-sagittal plane divides the body into bilaterally symmetric right and left halves. A coronal plane divides a body into anterior and posterior portions. A transverse plane divides a body into superior and inferior portions. Anterior means toward the front of the body. Posterior means toward the back of the body. Superior means toward the head. Inferior means toward the feet. Medial means toward the midline of the body. Lateral means away from the midline of the body. Axial means toward a central axis of the body. Abaxial means away from a central axis of the body. Ipsilateral means on the same side of the body. Contralateral means on the opposite side of the body. Proximal means located toward a center of a body, or toward a user. Distal means located away from the center of the body, or away from a user. These descriptive terms may be applied to an animate or inanimate body.

(166) Standard spinal descriptive terminology is used herein with the ordinary and customary

meanings.

(167) Referring to FIGS. 1 and 2, an instrument **100** for insertion and expansion of an intervertebral implant is disclosed. The instrument **100** extends from a proximal end **105** to a distal end **107** along a longitudinal instrument axis **102**. The instrument **100** includes a handle portion **108** and a shaft portion **110**. In FIG. 2, an expandable intervertebral implant **1000** is connected to the instrument distal end **107** and a draw bar assembly **160** is received in the instrument **100** for expansion of the implant.

(168) Referring to FIGS. 3-8, components and features of instrument **100** are disclosed. As seen in FIGS. 3 and 6, instrument **100** includes a lock collar **120** attachable to a shuttle **122**. A proximal lock tube **124** extends distally from the shuttle **122**, and a distal lock tube **126** extends distally from the proximal lock tube. A lock tube coupling **128** couples the proximal and distal lock tubes **124**, **126** together. The lock collar **120**, shuttle **122**, and proximal and distal lock tubes **124**, **126** are cannulated, surrounding a longitudinal instrument bore **112** extending along the longitudinal instrument axis **102**. As seen in FIGS. 3 and 7, a threaded driver **130**, standoff **132**, and standoff nut **134** are coupled to the shuttle **122**. An inserter shaft **136** extends distally from lock tube coupling **128**, and an inserter connector **138** is at the distal end of the inserter shaft **136**. The outer inserter shaft **136** may include one or more openings that aid in cleaning the instrument. As seen in FIGS. 3 and 8, the fully assembled instrument further includes an expansion knob **140**, inserter handle **142**, selector ring **144**, and impaction cap **146**. Markings or indicia **248** on the shuttle **122** or another component may provide indication when a particular step of implant expansion is attained. In one example the words “collapsed”, “lateral” and “vertical” may be indicated when the implant is in collapsed, laterally expanded, and vertically expanded configurations, respectively. In other embodiments, letters, numbers, symbols, or other markings may be used to indicate stages of insertion and/or expansion.

(169) Referring to FIGS. 9 and 10, the instrument **100** may include a shuttle assembly **114**, a standoff nut **134**, an inserter handle **142**, a selector ring **144**, an expansion knob **140**, a threaded driver **130**, a shaft assembly **116**, a standoff **132**, and a lock tube assembly **118**.

(170) Referring to FIGS. 11 and 12, the shuttle assembly **114** may include a shuttle **122** and a lock collar **120**.

(171) The shuttle **122** extends between a proximal end **184** and a distal end **186**. The proximal end may include a head **188**. A shaft **192** may extend from the head **188** toward the distal end **186**. The head **188** may be wider than the shaft **192**, as shown.

(172) The head **188** may include, from proximal to distal, first, second, third (FIG. 6), and fourth levels **194**, **196**, **198**, **202**. The first level **194** may have a generally rectangular cross section and may overhang the shaft **192** on three sides, as shown. The first level **194** may include a tab **204** that protrudes from a first side of the first level. The tab **204** may be flanked by bilateral teeth **206**, **208** that may also protrude from the first side of the first level. The teeth **206**, **208** may be smaller and/or shorter than the tab **204**. Bilateral indentations **210**, **212** may be present between the tab **204** and the teeth **206**, **208**. The second and third overhanging sides **214**, **216** of the first level **194** may be flat. A hole **234** may extend through the tab **204** and may be parallel to the shaft **192**. Bilateral holes **236**, **238** may extend into the first level **194** opposite the tab **204**. Hole **236** is visible in FIG. 12; hole **238** is hidden by the second level **196**. The second level **196** may have a generally rectangular cross section and may overhang the shaft **192** all around. The second level **196** may include the tab **204**, the teeth **206**, **208**, and the indentations **210**, **212** arranged on a first overhanging side of the second level. Second and third overhanging sides **218**, **220** may be flat, and may be parallel to the sides **214**, **216**. The side **218** may be recessed relative to the side **214** to form a ledge **222**. The side **220** may be recessed relative to the side **216** to form a ledge **224**. A fourth overhanging side **226** may include a shallow central indentation **228** opposite the tab **204**. The third level **198** (FIG. 6) may have a circular cross section and may include a circumferential groove **230**. The fourth level **202** may have a circular cross section and may include a circumferential flange

232.

(173) The shaft **192** may include, from proximal to distal, first, second, third, and fourth outer portions **240**, **242**, **244**, **246**. The first outer portion **240** may have a circular cross section and may include indicia **248** (FIG. 6) such as lines and/or text to indicate various implant configurations, such as collapsed, laterally expanded, and/or vertically expanded. The outer diameter of the first outer portion **240** may be equal to the minor diameter in the groove **230**. The second outer portion **242** may include external threads **250**. The major diameter of the threads **250** may be equal to the outer diameter of the first outer portion **240**. The third outer portion **244** may have a circular cross section. The outer diameter of the third outer portion **244** may be smaller than the minor diameter of the threads **250**. The fourth outer portion **246** may have a generally rectangular cross section. The cross section may be square with rounded corners as shown. Two opposite flat sides of the fourth outer portion **246** may be parallel to the sides **214**, **216** and/or the sides **218**, **220**. The width across the flat sides may be equal to the outer diameter of the third outer portion **244**. The width across the rounded corners may be greater than the outer diameter of the third outer portion **244** and may be less than the outer diameter of the first outer portion **240**.

(174) The shuttle **122** may include a central longitudinal hole **252** that extends through the shuttle between the proximal end **184** and the distal end **186**. The hole **252** may include, from proximal to distal, first, second, third, fourth (FIG. 5), and fifth inner portions **254**, **256**, **258**, **260**, **262**. Each of these inner portions may have a circular cross section. The first inner portion **254** may extend into the proximal end **184** through the first level **194** and may end at the second level **196**. The first inner portion **254** may be referred to as a counterbore associated with the second inner portion **256**. The first inner portion **254** may break through the side of the first level **194** opposite the tab **204**. The second inner portion **256** may extend through the second level **196**. The diameter of the second inner portion **256** may be less than the diameter of the first inner portion **254**. The third inner portion **258** may extend through the third and fourth levels **198**, **202** and most of the first outer portion **240**, and may end just proximal to the second outer portion **242**. There may be a short conical tapered transition between the second inner portion **256** and the third inner portion **258**. The diameter of the third inner portion **258** may be less than the diameter of the first inner portion **254** and may be greater than the diameter of the second inner portion **256**. The fourth inner portion **260** (FIG. 5) may extend between the third and fifth inner portions **258**, **262**. The fourth inner portion **260** may begin just proximal to the second outer portion **242** and may end just distal to the first outer portion **240**. There may be a short conical tapered transition between the third inner portion **258** and the fourth inner portion **260**. The diameter of the fourth inner portion **260** may be less than the diameter of the second inner portion **256**. The fifth inner portion **262** may extend into the distal end **186** through the fourth and third outer portions **246**, **244** and most of the second outer portion **242**, and may end just distal to the first outer portion **240**. The diameter of the fifth inner portion **262** may be less than the diameter of the first inner portion **254** and may be greater than the diameter of the second inner portion **256**. The diameter of the fifth inner portion **262** may be equal to the diameter of the third inner portion **258**.

(175) The lock collar **120** extends between a proximal side **264** and a distal side **266**. The lock collar **120** may have a generally D-shaped profile in a proximal view, with a curved side **268** and three substantially flat sides **270**, **272**, **274**. The proximal side **264** may be flat. A central hole **276** may extend through the lock collar **120** between the proximal side **264** and the distal side **266**. The hole **276** may be non-circular in a proximal view. The hole **276** may include a first curved side **278**, flat tapered sides **280**, **282**, flat parallel sides **284**, **286**, and a second curved side **288**. Transitional fillet surfaces may be present between some or all of the sides **278**, **280**, **282**, **284**, **286**, **288**. A conical surface **290** may be associated with the first curved side **278**, and may taper into the hole **276** from the proximal side **264**. A slot **292** may extend through the lock collar **120** from the proximal side **264** to the distal side **266**. The slot **292** may be located between the curved side **268** and the first curved side **278** and may be elongated in a direction from the curved side **268** to the

flat side **272**. The distal side **266** may include bilateral feet **294**, **296** that protrude distally on either side of the hole **276**. The foot **294** may be associated with the flat side **270**. The foot **296** may be associated with the flat side **274**. In a distal view, the outer profile of the curved side **268** may be concentric with the outer profiles of the feet **294**, **296**. The inner profiles of the feet **294**, **296** may be flat and parallel to each other, and may also be parallel to the flat sides **270**, **274**. The feet **294**, **296** may be undercut. An undercut groove **298** may extend along the inner profile of the foot **294** and an undercut groove **346** may extend along the inner profile of the foot **296**. Taken together, the feet **294**, **296** and undercut grooves **298**, **346** may form a T-slot across the distal side **266**. The distal side **266** may include a first surface **348** that is proximal to the distal aspect of the feet **294**, **296**. The first surface **348** may be at the same or a similar level as the distal-most surfaces of the undercut grooves **298**, **346**. The distal side **266** may include a second surface **350** that is proximal to the first surface **348**. The second surface **350** may be at the same or a similar level as the proximal-most surfaces of the undercut grooves **298**, **346**. Bilateral holes **352**, **354** (not shown) may extend into the lock collar, between the first surface **348** and the second surface **350**, along a direction from the curved side **268** to the flat side **272**. The bilateral holes **352**, **354** may be complementary to the bilateral holes **236**, **238** in the head **188** of the shuttle **122**. The flat side **272** may include friction features **356**, such as alternating ridges and grooves, knurling, grit blast, or the like.

(176) The shuttle assembly **114** may be assembled by orienting the lock collar **120** relative to the head **188** of the shuttle **122** so that the proximal end **184** and the proximal side **264** face the same way, the side **226** and the side **272** face the same way, and the sides **218**, **220** are aligned between the feet **294**, **296**; and sliding the lock collar **120** along a direction from the flat side **272** to the curved side **268** so that the first level **194** is received in the undercut grooves **298**, **346** and the second level **196** is received between the feet **294**, **296**. Bilateral pins or compression springs (not shown) may be inserted into the bilateral holes **236**, **238** and **352**, **354**. A pin (not shown) may be inserted through the slot **292** and into the hole **234**. When compression springs are inserted in the holes **236**, **238** and **352**, **354** and a pin is inserted through the slot **292** and into the hole **234**, the springs bias the lock collar **120** toward a position in which the pin contacts the end of the slot **292** that is closest to the outer curved side **268**.

(177) Referring to FIGS. **9** and **10**, the standoff nut **134** extends between a proximal side **358** and a distal side **360**. The standoff nut **134** may be a generally cylindrical part. The proximal side **358** may include a circumferential outer chamfer **362** and/or bilateral notches **364**, **366**. Each notch may include a longitudinal surface **368** and a transverse surface **370**. The distal side **360** may include a circumferential outer chamfer **372**. A central hole **374** may extend through the standoff nut **134** between the proximal side **358** and the distal side **360**. The hole **374** may include, from proximal to distal, first, second, and third inner portions **376**, **378**, **380**. The first inner portion **376** may have a circular cross section. The second inner portion **378** may have a circular cross section, and may have a larger diameter than the first portion **376**. The second inner portion **378** may have a flat proximal end. The third inner portion **380** may have internal threads, which may have a major diameter the same or similar to the diameter of the second portion **378**. The hole **374** may include an inner chamfer **382** at the distal side **360**.

(178) Referring to FIGS. **5**, **9**, **10**, **13**, and **14**, the inserter handle **142** extends between a proximal end **384** and a distal end **386**. The inserter handle **142** may be generally circular in cross section. The inserter handle **142** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth outer portions **388**, **390**, **392**, **394**, **396**, **398**. The first outer portion **388** may be cylindrical with external threads. The second outer portion **390** may be cylindrical with a diameter that may be the same or similar to the minor diameter of the external threads of the first outer portion **388**. The third outer portion **392** may be cylindrical with a diameter that may be greater than the major diameter of the external threads of the first outer portion **388**. The fourth outer portion **394** may be generally hourglass-shaped and may include friction features **400**, such as longitudinal grooves,

alternating ridges and grooves, knurling, grit blast or the like. The fifth outer portion **396** may be cylindrical with a diameter that may be greater than the diameter of the third outer portion **392**. The fifth outer portion **396** may include indicia **402**, such as a distally-pointing arrow as shown, and/or an outer chamfer **404**. The sixth outer portion **398** may be cylindrical with a diameter that may be the same or similar to the diameter of the second outer portion **390**. The sixth outer portion **398** may include a short base portion **406** which may be directly adjacent to the fifth outer portion **396**, and a collar portion **408** which is at the distal end **386**. The collar portion **408** may be horseshoe-shaped or C-shaped in cross section, with an open side that faces the distally-pointing arrow indicium **402**. Bilateral holes **410**, **412** may extend proximally into the ends of the collar portion **408** (the ends of the C-shape).

(179) The inserter handle **142** may include a central longitudinal hole **414** that extends through the inserter handle between the proximal end **384** and the distal end **386**. The hole **414** may include, from proximal to distal, first, second, third, fourth, and fifth inner portions **416**, **418**, **420**, **422**, **424**. The first inner portion **416** may have a circular cross section and may extend distally through the first and second outer portions **388**, **390** and most of the third outer portion **392**, and may end proximal to the fourth outer portion **394**. The second inner portion **418** may have a generally rectangular cross section. The cross section may be square with rounded corners as shown, and the open side of the collar portion **408** may face one of the flat sides. The width across the flat sides and the width across the rounded corners may each be less than the diameter of the first inner portion **416**. The second inner portion **418** may extend to a location that is near the middle of the overall proximal-distal length of the inserter handle **142**. The third inner portion **420** may have a circular cross section. The diameter of the third inner portion **420** may be the same or similar to the width across the flat sides of the second inner portion **418**. The third inner portion **420** may end just proximal to the sixth outer portion **398**. The fourth inner portion **422** may have a circular cross section. The diameter of the fourth inner portion **422** may be less than the width across the flat sides of the second inner portion **418** or the diameter of the third inner portion **420**. Bilateral longitudinal grooves **426**, **428** may extend along the fourth inner portion **422**. The width across the grooves **426**, **428** may be the same or similar to the width across the flat sides of the second inner portion **418** or the diameter of the third inner portion **420**. The groove **428** may point into the open side of the collar portion **408**, near the end with the hole **412**. The grooves **426**, **428** may point into the rounded corners of the second inner portion **418**. The fourth inner portion **422** may end just distal to the fifth outer portion **396**, or near the middle of the overall proximal-distal length of the base portion **406**. The fifth inner portion **424** may have a circular cross section and may be continuous with the inner wall of the collar portion **408**. The diameter of the fifth inner portion **424** may be less than the width across the rounded corners of the second inner portion **418**, greater than the width across the flat sides of the second inner portion or the diameter of the third inner portion **420**, and greater than the width across the grooves **426**, **428**. The fifth inner portion **424** may be referred to as a shallow counterbore associated with the fourth inner portion **422**.

(180) Referring to FIGS. **9**, **10**, and **15**, the selector ring **144** extends between a proximal side **430** and a distal side **432**. The selector ring **144** may include an outer ring **434**, an inner ring **436** within the outer ring, and an arm **438** that connects the inner ring to the outer ring. The outer ring **434** may be a generally cylindrical part with flat proximal and distal sides **430**, **432**. The outer surface of the outer ring **434** may include a longitudinal groove **440** and a longitudinal rib **442**. The illustrated example includes four longitudinal grooves **440**, **444**, **446**, **448** and three longitudinal ribs **442**, **450**, **452**. The outer surface of the outer ring **434** may also include indicia **454** such as lines, text, and/or icons as shown to indicate various settings, such as a locked setting, an unlocked setting, and a setting for cleaning. The inner surface of the outer ring **434** may be cylindrical. The inner ring **436** may be a generally cylindrical part with flat proximal and distal sides **456**, **458**. The proximal side **456** may be recessed relative to the proximal side **430** and the distal side **458** may be recessed relative to the distal side **432**. The outer surface of the inner ring **436** may be cylindrical as shown.

The inner surface of the inner ring **436** may be cylindrical with bilateral longitudinal rectangular grooves **460**, **462**. The arm **438** may extend between the inner surface of the outer ring **434** and the outer surface of the inner ring **436**. The arm **438** may be aligned with the groove **462**. The arm **438** may have a longitudinal hole **464** that extends between the proximal and distal sides **456**, **458**. (181) Referring to FIGS. **9**, **10**, **16**, and **17**, the expansion knob **140** extends between a proximal end **466** and a distal end **468**. The expansion knob **140** may be generally circular in cross section. The expansion knob **140** may be wider toward the proximal end **466** and narrower toward the distal end **468**. The expansion knob **140** may include friction features **470**, such as longitudinal grooves, alternating ridges and grooves, knurling, grit blast, or the like. Bilateral tabs **472**, **474** may extend proximally. The tabs **472**, **474** may have circular outer and inner profiles. The outer sides of the tabs **472**, **474** may include indicia **476**, such as transverse lines as shown. Bilateral holes **478**, **480** may extend distally into the expansion knob **140** and may be located between the tabs **472**, **474** (i.e., rotated 90 degrees away from the tabs). The holes **478**, **480** may include proximal counterbores.

(182) The expansion knob **140** may include a central longitudinal hole **482** that extends through the expansion knob between the proximal end **466** and the distal end **468**. The hole **482** may include, from proximal to distal, first, second, third, and fourth inner portions **484**, **486**, **488**, **490**. The first inner portion **484** may have a circular cross section. The first inner portion **484** may extend distally about one third of the overall proximal-distal length of the expansion knob **140**. The second inner portion **486** may have a generally rectangular cross section. The cross section may be square with rounded corners as shown. The width across the flat sides and the width across the rounded corners may each be greater than the diameter of the first inner portion **484**. The second inner portion **486** may extend distally about one third of the overall proximal-distal length of the expansion knob **140**, in other words, the second inner portion may occupy the middle third of the overall proximal-distal length of the expansion knob. The third inner portion **488** may have a circular cross section. The diameter of the third inner portion **488** may be greater than the width across the rounded corners of the second inner portion **486**. The third inner portion **488** may end just proximal to the distal end **468**. The fourth inner portion **490** may have a circular cross section. The diameter of the fourth inner portion **490** may be the same, similar to, or greater than the width across the rounded corners of the second inner portion **486** and less than the diameter of the third inner portion **488**. The fourth inner portion **490** may form a shallow undercut with the third inner portion **488**.

(183) Referring to FIGS. **9**, **10**, **18**, and **19**, the threaded driver **130** extends between a proximal end **492** and a distal end **494**. The threaded driver **130** may be referred to as an actuator. The threaded driver **130** may be a rectangular solid, and may be a cube. Referring to FIGS. **18** and **19**, the threaded driver **130** may have rounded corners in a proximal or distal view. Bilateral holes **496**, **498** may extend into the proximal end **492** about a third of the overall proximal-distal length of the threaded driver **130**. The holes **496**, **498** may be located in opposite corners of the proximal end **492**. The proximal ends of the holes **496**, **498** may include shallow countersinks. The threaded driver **130** may include a central longitudinal hole **500** that extends through the threaded driver between the proximal end **492** and the distal end **494**. The hole **500** may include, from proximal to distal, first, second, third, fourth, and fifth inner portions **502**, **504**, **506**, **508**, **510**. The first inner portion **502** may be a shallow conical countersink around the hole **500** in the proximal end **492**. The second inner portion **504** may be a shallow cylindrical counterbore. The third inner portion **506** may include internal threads **512** that receive the external threads **250** of the shuttle **122**. Referring to FIGS. **18** and **19**, the cross-sectional shape of the third inner portion **506** may be the superposition of a circle defining the minor diameter of the internal threads **512** and a rectangle. The minor diameter of the internal threads **512** may be smaller than the diameter of the second inner portion **504**. The major diameter of the internal threads **512** may be greater than the diameter of the second inner portion **504**. The rectangle may be a square, and may have rounded corners. The width across the flat sides may be slightly less than the minor diameter of the internal threads

512. The width across the corners may be slightly less than the major diameter of the internal threads **512**. The fourth inner portion **508** may be a shallow cylindrical counterbore, with a diameter greater than the minor diameter of the internal threads **512** and less than the major diameter of the internal threads **512**. The diameter of the fourth inner portion **508** may be equal to, or similar to, the diameter of the second inner portion **504**. The fifth inner portion **510** may be a shallow counterbore around the hole **500** in the distal end **494**, with a diameter greater than the major diameter of the internal threads **512**.

(184) Referring to FIGS. **9**, **10**, **22**, and **23**, the shaft assembly **116** may include an impaction cap **146**, an inserter connector **138**, and an inserter shaft **136**.

(185) The impaction cap **146** extends between a proximal side **514** and a distal side **516**. The impaction cap **146** may be circular in a proximal or distal view. The impaction cap **146** may include, from proximal to distal, first, second, and third outer portions **518**, **520**, **522**. The first outer portion **518** may be cylindrical. The second outer portion **520** may be conical, and may be referred to as a chamfer. The third outer portion **522** may be cylindrical, and may have a diameter greater than the diameter of the first outer portion **518**. An outer distal edge of the third outer portion **522** may be rounded. Indicia **524**, such as the mark **524** shown in FIG. **22**, may be included on the third outer portion **522**. One or more dimples **526** may be included in the proximal side; three evenly spaced dimples **526**, **528**, **530** are shown. The middle dimple **528** may be aligned with the mark **524**. Bilateral holes **532**, **534** may extend through the impaction cap **146** between the proximal and distal sides **514**, **516**. The holes **532**, **534** may be symmetrically located relative to the mark **524**. Referring to FIG. **23**, each hole **532**, **534** may include a conical distal countersink. A central hole **536** may extend through the impaction cap **146** between the proximal and distal sides **514**, **516**.

(186) The inserter connector **138** extends between a proximal end **538** and a distal end **540**. The proximal end **538** may be circular in a proximal view. A notch **542** may be present in an outer proximal edge of the proximal end **538**. The notch **542** may be rounded or semicircular in a side view. The distal end **540** may be rectangular in a distal view. The distal end **540** may have rounded corners in the distal view. The major dimension (length) of the rectangle may be aligned with the notch **542**. The inserter connector **138** may include a central longitudinal hole **544** that extends through the inserter connector between the proximal end **538** and the distal end **540**. The hole **544** may include, from proximal to distal, first, second, and third inner portions **546**, **548**, **550**. The first inner portion **546** may have a circular cross section and may extend distally about one third of the overall proximal-distal length of the inserter connector **138**. The first inner portion **546** may have a flat distal end. The second inner portion **548** may have a circular cross section with a diameter less than the first inner portion **546**. The second inner portion **548** may extend distally about one third of the overall proximal-distal length of the inserter connector **138**. Referring to FIG. **23**, the cross-sectional shape of the third inner portion **550** may be the superposition of a circle and a rectangle. The diameter of the circle may be less than the diameter of the first inner portion **546** and greater than the diameter of the second inner portion **548**. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion **546**, and may be aligned with the major dimension (length) of the rectangle of the distal end **540**. The minor dimension (width) of the rectangle may be less than the diameter of the circle, and may be less than the diameter of the second inner portion **548**. The rectangle may have rounded corners. The third inner portion **550** may have a flat proximal end. An inner distal edge of the rectangle may be chamfered.

(187) The inserter shaft **136** extends between a proximal end **552** and a distal end **554**. The inserter shaft **136** may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft **136** may include, from proximal to distal, first, second, and third outer portions **556**, **558**, **560**. The first outer portion **556** may include a pair of shaped slots **562**, **564** which each include an approximately 90° bend. The slots **562**, **564** may extend longitudinally into the proximal end **552** before making the approximately 90° bend. Each slot **562**, **564** may extend through one wall of the

first outer portion **556**. The outer diameter of the second outer portion **558** may be greater than the outer diameter of the first outer portion **556**. The second outer portion **558** may occupy most of the overall proximal-distal length of the inserter shaft **136**. The second outer portion **558** may include one or more openings **566**. Four openings **566** are shown evenly spaced along the length of the second outer portion **558**. Each opening **566** may extend through one wall of the second outer portion **558**, or through both walls as shown. The second outer portion **558** may include one or more indicia or markings. Three indicia are shown, from proximal to distal: a sideways “8” **568**, a longitudinal line **570**, and a set of numerals and transverse lines **572**. The indicia may be repeated on the opposite side of the second outer portion **558** as shown in FIGS. **22** and **23**. The outer diameter of the third outer portion **560** may be less than the outer diameter of the first outer portion **556**. The inserter shaft **136** may include a central longitudinal hole **574** that extends through the inserter shaft between the proximal end **552** and the distal end **554**. The hole **574** may be chamfered around its proximal inner edge. The diameter of the hole **574** may be the same as, or similar to, the diameter of the second inner portion **548** of the inserter connector.

(188) The shaft assembly **116** may be assembled by inserting the first outer portion **556** of the inserter shaft **136** into the central hole **536** of the impaction cap **146** so that the proximal side **514** and the proximal end **552** face the same direction, the distal side **516** abuts the proximal end of the second outer portion **558**, and the mark **524** is aligned with the sideways “8” **568** and/or the longitudinal line **570**; and by inserting the third outer portion **560** of the inserter shaft **136** into the first inner portion **546** of the inserter connector **138** so that the major dimension of the rectangle of the distal end **540** is transverse to, or perpendicular to, the sideways “8” **568** and/or the longitudinal line **570**. The shaft assembly **116** may be a weldment. Alternatively, the impaction cap **146**, inserter connector **138**, and/or inserter shaft **136** may be combined into a single part.

(189) Referring to FIGS. **9** and **10**, the standoff **132** extends between a proximal end **576** and a distal end **578**. The standoff **132** may be a tubular part with circular outer and inner cross-sectional shapes. The standoff **132** may include, from proximal to distal, first and second outer portions **580**, **582**. The first outer portion **580** may extend over most of the overall proximal-distal length of the standoff **132**. The outer diameter of the second outer portion **580** may be greater than the outer diameter of the first outer portion **580** so that the second outer portion **580** forms a flange around the distal end **578**. The standoff **132** may include a central longitudinal hole **584** that extends through the standoff between the proximal end **576** and the distal end **578**.

(190) Referring to FIGS. **9**, **10**, **24**, and **25**, the lock tube assembly **118** may include a proximal lock tube **124**, a proximal lock tube end part **148**, a distal lock tube **126**, and a lock tube coupling **128**.

(191) The proximal lock tube **124** extends between a proximal end **586** and a distal end **588**. The proximal lock tube **124** may be a tubular part with circular outer and inner cross-sectional shapes. The proximal lock tube **124** may include, from proximal to distal, first and second outer portions **590**, **592**. The first outer portion **590** may extend distally a short distance. The diameter of the second outer portion **592** may be greater than the diameter of the first outer portion **590**. The second outer portion **592** may extend over most of the overall proximal-distal length of the proximal lock tube **124**. The proximal lock tube **124** may include a central longitudinal hole **594** that extends through the proximal lock tube between the proximal end **586** and the distal end **588**. The hole **594** may be chamfered around its proximal inner edge.

(192) The proximal lock tube end part **148** extends between a proximal end **596** and a distal end **598**. The proximal lock tube end part **148** may be a tubular part with circular outer and inner cross-sectional shapes. The proximal lock tube end part **148** may include, from proximal to distal, first, second, and third outer portions **600**, **602**, **604**. The first outer portion **600** may extend distally a short distance, and may include a chamfer around its proximal outer edge. The diameter of the second outer portion **602** may be less than the diameter of the first outer portion **600**. The second outer portion **602** may also extend distally a short distance. The diameter of the third outer portion **604** may be greater than the diameter of the second outer portion **602**, and may be equal to the

diameter of the first outer portion **600**. Thus, the second outer portion **602** may be referred to as an exterior groove around the proximal end **596**. There may be a short tapered transition between the distal end of the second outer portion **602** and the proximal end of the third outer portion **604**. The proximal lock tube end part **148** may include a central longitudinal hole **606** that extends through the proximal lock tube end part between the proximal end **596** and the distal end **598**. The hole **606** may include a chamfer **608** around its proximal inner edge, and may include a distal counterbore **610**. The inner diameter of the hole **606** may be the same as, or similar to, the inner diameter of the hole **594**.

(193) The distal lock tube **126** extends between a proximal end **612** and a distal end **614**. The distal lock tube **126** may be a tubular part with circular outer and inner cross-sectional shapes. The proximal end **612** may include a transverse notch **616** that extends through both walls of the distal lock tube **126**. The notch **616** may include a rounded distal end. The notch **616** may divide the proximal end **612** into a pair of tabs **618**, **620**. The notch **616** may step up in width proximally and correspondingly, each tab **618**, **620** may step down in width proximally to form fingers **622**, **624**. The distal end **614** may be chamfered around its outer distal edge. With brief reference to FIG. **49**, the distal end **614** may include a pair of shaped slots **152** which each include an approximately 90° bend. The slots **152** may extend longitudinally into the distal end **614** before making the approximately 90° bend. Each slot **152** may extend through one wall of the distal lock tube **126**. The distal lock tube **126** may include a central longitudinal hole **626** that extends through the distal lock tube between the proximal end **612** and the distal end **614**. The inner diameter of the hole **626** may be the same as, or similar to, the inner diameter of the hole **594**.

(194) Referring to FIGS. **24**, **25**, **20**, and **21**, the lock tube coupling **128** extends between a proximal end **628** and a distal end **630**. The lock tube coupling **128** may be a tubular part. The lock tube coupling **128** may include a proximal body **632** which may have a circular cross-sectional outer shape, and may extend distally about half or three-fifths of the overall proximal-distal length of the lock tube coupling **128**. Bilateral tabs **634**, **636** may protrude distally from opposite sides of the body **632**. The tabs **634**, **636** may be inset relative to the outer diameter of the body **632**. The distal portions of the tabs **634**, **636** may include posts **638**, **640**, respectively. The posts **638**, **640** may protrude outwardly from the outer sides of the tabs **634**, **636**, and may protrude beyond the outer diameter of the body **632** as shown in FIGS. **20** and **21**. The lock tube coupling **128** may include a central longitudinal hole **642** that extends through the lock tube coupling between the proximal end **628** and the distal end **630**. The hole **642** may include, from proximal to distal, first, second, and third inner portions **644**, **646**, **648**. The first inner portion **644** may have a circular cross section and may extend distally about a third of the overall proximal-distal length of the lock tube coupling **128**, or about half of the length of the body **632**. The second inner portion **646** may have a circular cross section with a diameter less than the diameter of the first inner portion **644**. The diameter of the second inner portion **646** may be the same as, or similar to, the inner diameter of the hole **594**. The second inner portion **646** may extend distally only a short distance as a complete diameter, but sections of it may extend through the distal end to form the inner sides of the tabs **634**, **636**. The third inner portion **648** may have a non-circular cross section that includes bilateral alcoves **650**, **652** in the distal walls of the body **632** between the tabs **634**, **636** (in other words, the alcoves **650**, **652** are located 90° away from the tabs **634**, **636**).

(195) The lock tube assembly **118** may be assembled by inserting the first outer portion **590** of the proximal lock tube **124** into the distal counterbore **610** of the proximal lock tube end part **148**; inserting the distal end **588** of the proximal lock tube **124** into the first inner portion **644** of the lock tube coupling **128**; and inserting the proximal end **612** of the distal lock tube **126** into the third inner portion **648** of the lock tube coupling **128** so that the fingers **622**, **624** are received in the alcoves **650**, **652** and the tabs **634**, **636** are received in the notch **616**. The hole **594** of the proximal lock tube **124**, the hole **606** of the proximal lock tube end part **148**, the hole **626** of the distal lock tube **126**, and the second inner portion **646** of the hole **642** of the lock tube coupling **128** may

together form the longitudinal instrument bore **112**. The proximal lock tube **124**, proximal lock tube end part **148**, distal lock tube **126**, and lock tube coupling **128** may be permanently connected, for example by welding. Alternatively, the proximal lock tube **124**, proximal lock tube end part **148**, distal lock tube **126**, and/or lock tube coupling **128** may be combined into a single part.

(196) The instrument **100** may be assembled by coupling the shaft assembly **116** to the lock tube assembly **118**; coupling the selector ring **144** to the shaft assembly **116** and lock tube assembly **118**; coupling the inserter handle **142** to the shaft assembly **116**, lock tube assembly **118**, and selector ring **144**; coupling the standoff **132** and the standoff nut **134** to the shaft assembly **116**, lock tube assembly **118**, selector ring **144**, and inserter handle **142**; coupling the threaded driver **130** and the shuttle assembly **114** to the expansion knob **140**; coupling the shuttle assembly, threaded driver, and expansion knob **140** to the shaft assembly **116**, lock tube assembly **118**, selector ring **144**, and inserter handle **142**.

(197) Coupling the shaft assembly **116** to the lock tube assembly **118** may include inserting the distal end **614** of the distal lock tube **126** into the hole **574** in the proximal end **552** of the inserter shaft **136**, advancing the lock tube assembly **118** distally until the posts **638**, **640** enter the shaped slots **562**, **564**, and twisting the lock tube assembly **118** counterclockwise so that the posts **638**, **640** are captured in the shaped slots **562**, **564**.

(198) Coupling the selector ring **144** to the shaft assembly **116** and lock tube assembly **118** may include inserting the proximal end **596** of the proximal lock tube end part **148** into the inner ring **436** of the selector ring **144** and advancing the lock tube assembly **118** proximally until the posts **638**, **640** enter the grooves **460**, **462** and the hole **464** is aligned with the middle dimple **528**. A compression spring or a ball detent may be received in the hole **464** to engage the dimples **526**, **528**, **530**.

(199) Coupling the inserter handle **142** to the shaft assembly **116**, lock tube assembly **118**, and selector ring **144** may include inserting the proximal end **596** of the proximal lock tube end part **148** into the hole **414** in the distal end **386** of the inserter handle **142**, advancing the lock tube assembly **118** proximally until the collar portion **408** enters the space between the outer ring **434** and the inner ring **436**, and inserting fasteners such as screws through the holes **532**, **410** and the holes **534**, **412**.

(200) Coupling the standoff **132** and the standoff nut **134** to the shaft assembly **116**, lock tube assembly **118**, selector ring **144**, and inserter handle **142** may include inserting the proximal end **596** of the proximal lock tube end part **148** into the hole **584** in the distal end **578** of the standoff **132**, advancing the lock tube assembly **118** proximally until the distal end **578** abuts the proximal end **384** of the inserter handle **142**, inserting the proximal end **596** into the hole **374** in the distal side **360** of the standoff nut **134**, advancing the lock tube assembly **118** proximally until the distal side **360** abuts the proximal end **384**, and engaging the internal threads of the third inner portion **380** with the external threads of the first outer portion **388** to make a tight connection. The notches **364**, **366** may facilitate tightening.

(201) Coupling the threaded driver **130** and the shuttle assembly **114** to the expansion knob **140** may include inserting the proximal end **492** of the threaded driver **130** into the hole **482** in the distal end **468** of the expansion knob **140**, aligning the hole **478** with the hole **496**, advancing the threaded driver **130** proximally into the second inner portion until the proximal end **492** abuts the proximal end of the second inner portion **486**, inserting the distal end **186** of the shuttle **122** into the hole **482** in the proximal end **466** of the expansion knob **140**, advancing the shuttle assembly **114** distally until the distal end **186** abuts the proximal end **492** of the threaded driver **130**, aligning the fourth outer portion **246** with the third inner portion **506**, advancing the shuttle assembly **114** farther distally until the distal end of the second outer portion **242** abuts the proximal end **492**, and engaging the external threads **250** of the second outer portion **242** with the internal threads **512** of the third inner portion **506** until the shuttle assembly **114**, expansion knob **140**, and threaded driver **130** achieve a snug connection.

(202) Coupling the shuttle assembly **114**, threaded driver **130**, and expansion knob **140** to the shaft assembly **116**, lock tube assembly **118**, selector ring **144**, and inserter handle **142** may include inserting the proximal end **596** of the proximal lock tube end part **148** into the hole **252** in the distal end **186** of the shuttle; advancing the lock tube assembly **118** proximally until the fifth inner portion **510** of the threaded driver **130** receives the proximal end **576** of the standoff **132**, the third inner portion **488** of the expansion knob **140** receives the standoff nut **134**, the fourth inner portion **490** of the expansion knob **140** receives the third outer portion **392** of the inserter handle **142**, the distal end **468** of the expansion knob **140** abuts the distal end of the third outer portion **392**, and the second inner portion **418** of the hole **414** of the inserter handle **142** receives the fourth outer portion **246** of the **122**; and inserting fasteners such as screws through the holes **478**, **496** and the holes **480**, **498**.

(203) The instrument **100** may be operated by sliding the lock collar **120**, rotating the selector ring **144**, or rotating the expansion knob **140**.

(204) Referring to FIGS. **11** and **12**, the lock collar **120** is coupled to the shuttle **122** so that the lock collar **120** can slide relative to the shuttle **122** along a direction established by the undercut grooves **298**, **346**, feet **294**, **296**, and slot **292**. The travel of the lock collar **120** may be limited by the presence of a pin in the slot **292** and the hole **234** of the shuttle **122**. In this arrangement, the lock collar **120** may slide between a first position and a second position. In the first position, the pin may contact the end of the slot **292** that is closest to the first curved side **278** of the hole **276**. The first position may be referred to as an open or unlocked position. In the second position, the pin may contact the end of the slot **292** that is closest to the outer curved side **268**. The second position may be referred to as a closed or locked position. The lock collar **120** may be biased toward the second position.

(205) Referring to FIGS. **9**, **10**, and **15**, the selector ring **144** is coupled to the posts **638**, **640** of the lock tube coupling **128** so that rotation of the selector ring **144** causes rotation of the lock tube assembly **118** relative to the rest of the instrument **100**. Referring to FIGS. **14** and **15**, rotation of the selector ring **144** is limited by the presence of the collar portion **408** in the space between the outer ring **434** and the inner ring **436**. In the example shown, the selector ring **144** is rotatable between a first position and a third position. In the first position, the arm **438** may contact the end of the collar portion **408** that includes the hole **412**, the hole **464** may align with the dimple **530**, the indicia **454** for the cleaning setting may align with the distally-pointing arrow indicium **402**, the posts **638**, **640** may be positioned in the shaped slots **562**, **564** so that the posts are free to slide axially out of the slots, and the posts **638**, **640** may be aligned with the grooves **426**, **428**. The first position may be referred to as a cleaning position because the lock tube assembly **118** and/or other parts of the instrument **100** are positioned for disassembly for cleaning or reassembly after cleaning. In the third position, the arm **438** may contact the end of the collar portion **408** that includes the hole **410**, the hole **464** may align with the dimple **526**, the indicia **454** for the locked setting may align with the distally-pointing arrow indicium **402**, the posts **638**, **640** may be positioned in the shaped slots **562**, **564** so that the posts are captured in the slots, the posts **638**, **640** may be rotated out of alignment with the grooves **426**, **428**, and the shaped slots **152** may be positioned with their distal openings aimed into the corners of the third inner portion **550** of the hole **544** of the inserter connector **138**. The third position may be referred to as a locked position because the parts of the instrument **100** are positioned to lock the implant **1000** to the instrument. The selector ring **144** may be rotatable to a second position which is between the first and third positions. In the second position, the arm **438** may be centered in the open side of the collar portion **408**, the hole **464** may align with the dimple **528**, the indicia **454** for the unlocked setting may align with the distally-pointing arrow indicium **402**, the posts **638**, **640** may be positioned in the shaped slots **562**, **564** so that the posts are captured in the slots, the posts **638**, **640** may be rotated out of alignment with the grooves **426**, **428**, and the shaped slots **152** may be positioned with their distal openings aimed along the major dimension (length) of the third inner portion **550** of the hole **544** of

the inserter connector **138**. The second position may be referred to as an unlocked position because the parts of the instrument **100** are positioned so that the implant **1000** may be coupled to or decoupled from the instrument **100**.

(206) Referring to FIGS. **9-12**, **13**, **14**, and **16-19**, the expansion knob **140** is coupled to the shuttle assembly **114**, the threaded driver **130**, and the inserter handle **142** so that rotation of the expansion knob **140** causes linear translation of the shuttle assembly **114** along the longitudinal instrument axis **102** without rotation of the shuttle assembly **114** about the axis **102**. The threaded driver **130** is received in the second inner portion **486** of the hole **482** of the expansion knob **140** so that rotation of the expansion knob **140** causes rotation of the threaded driver **130**. The external threads **250** of the second outer portion **242** of the shuttle **122** are received by the internal threads **512** of the third inner portion **506** of the threaded driver **130** and the fourth outer portion **246** of the shuttle **122** is received in the second inner portion **418** of the hole **414** of the inserter handle **142** so that rotation of the threaded driver **130** causes linear translation of the shuttle assembly **114** along the longitudinal instrument axis **102**. In the example shown, the shuttle assembly **114** is movable between a first position and a third position. In the first position, the distal side of the flange **232** of the shuttle **122** may abut the proximal end **466** (tabs **472**, **474**) of the expansion knob **140**, the proximal end of the external threads **250** of the second outer portion of the shuttle **122** may be even with, or close to, the proximal end **492** of the threaded driver **130**, and the indicia **248** of the shuttle **122** for the collapsed configuration may be aligned with the indicia **476** of the expansion knob **140**. The first position may be referred to as a collapsed position for this reason. In the third position, the distal end of the external threads **250** may be even with, or close to, the distal end **494** of the threaded driver **130**, and the indicia **248** of the shuttle **122** for the vertically expanded configuration may be aligned with the indicia **476** of the expansion knob **140**. The third position may be referred to as a vertically expanded position for this reason. The shuttle assembly **114** may be movable to a second position which is between the first and third positions. In the second position, the distal end of the external threads **250** may be even with, or close to, the proximal end **358** of the standoff nut **134** and the indicia **248** of the shuttle **122** for the laterally expanded configuration may be aligned with the indicia **476** of the expansion knob **140**. The second position may be referred to as a laterally expanded position for this reason. The shuttle assembly **114** may be movable to any number of positions between the first and third positions due to the threaded interconnection between the shuttle **122** and the threaded driver **130**.

(207) Various additional tools may be used with instrument **100** to implant and expand an intervertebral implant. The instrument **100** and selected additional tools may be provided in a kit and may be collectively referred to as an instrument system **115**. FIGS. **26-30** disclose additional tools of the instrument system **115**, including a draw bar assembly **160**, a graft funnel **170**, a tamp assembly **180**, and a screwdriver assembly **190**. The draw bar assembly **160** may include a threaded tip **162** and a handle **164**. The screwdriver assembly **190** may be assembled with a screw holder **200** to hold and drive a lockout screw **654** into an implant such as implant **1000** to lock the implant in an expanded configuration. The instrument system **115** may be used to implant and/or expand other expandable implants.

(208) Referring to FIGS. **26**, **31**, and **32**, the draw bar assembly **160** may include a draw bar **156** and a draw bar knob or handle **164**.

(209) The draw bar **156** extends between a proximal end **656** and a distal end **658**. The draw bar **156** may be a cylindrical shaft as shown. The draw bar **156** may include, from proximal to distal, first, second, and third outer portions **660**, **662**, **664**. The first outer portion **660** may extend distally most of the overall proximal-distal length of the draw bar **156**, and may be chamfered around its proximal outer edge. The diameter of the second outer portion **662** may be less than the diameter of the first outer portion **660**. A tapered shoulder **668** may be included between the first and second outer portions **660**, **662**. The third outer portion **664** may include the threaded tip **162** with external threads. The major diameter of the threads may be less than the diameter of the second outer

portion **662**.

(210) The draw bar handle **164** extends between a proximal end **670** and a distal end **672**. The draw bar handle **164** may include, from proximal to distal, a knob body **674** and a shaft **676**. The knob body **674** may be generally disc-shaped with an outer diameter that is greater than its proximal-distal length. The knob body **674** may include one or more grip features **678** such as the six rim scallops shown. The outer diameter of the shaft **676** may be less than the outer diameter of the knob body **674** and the proximal-distal length of the shaft may be greater than the proximal-distal length of the knob body **674**. The shaft **676** may include a flange **680** (annulus **166**) that extends circumferentially around a middle portion of the shaft. The flange **680** may include a distal chamfer **682**. The shaft **676** may include one or more holes **684** that extend transversely into, or through, the shaft; two through holes **684** are shown just distal to the distal side of the knob body **674**. The two through holes **684** are perpendicular to each other. The draw bar handle **164** may include a central longitudinal hole **686** that extends through the draw bar handle between the proximal end **670** and the distal end **672**. The hole **686** may include, from proximal to distal, first, second, third, and fourth inner portions **688**, **690**, **692**, **694**. The first inner portion **688** may have a circular cross section and may extend distally about half, or most, of the proximal-distal length of the knob body **674**. The first inner portion **688** may have a flat distal end. A circumferential fillet radius may be present around the proximal inner edge of the first inner portion **688**. A circumferential fillet radius may be present in the distal inner corner of the first inner portion **688**. The second inner portion **690** may be conical. The second inner portion **690** may be referred to as a countersink or circumferential chamfer associated with the third inner portion **692**. The major diameter of the second inner portion **690** may be less than the diameter of the first inner portion **688**. The third inner portion **692** may have a circular cross section and may extend distally to the same or similar level as the distal side of the knob body **674**. The diameter of the third inner portion **692** may be less than the diameter of the first inner portion **688**. The fourth inner portion **694** may have a circular cross section and may extend distally through the distal end of the **672** of the draw bar handle **164**. The diameter of the fourth inner portion **694** may be less than the diameter of the first inner portion **688** and greater than the major diameter of the second inner portion **690** and the diameter of the third inner portion **692**. The fourth inner portion **694** may have a flat proximal end. The hole(s) **684** may extend into a proximal end of the fourth inner portion **694**.

(211) The draw bar assembly **160** may be assembled by inserting the proximal end **656** of the draw bar **156** into the hole **686** of the draw bar handle **164** so that the first portion **660** is received in the third inner portion **692** and the proximal end **656** is at the same or similar level with the distal end of the first inner portion **688**. A circumferential gap **696** may exist between the outer surface of the first portion **660** and the inner surface of the fourth inner portion **694**. See FIG. 5. The draw bar **156** and draw bar handle **164** may be permanently connected, for example by welding. Alternatively, the draw bar **156** and draw bar handle **164** may be combined into a single part.

(212) The draw bar **156** may be received in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

(213) Referring to FIGS. **28**, **33**, and **34**, the graft funnel **170** extends between a proximal end **698** and distal end **700**. The graft funnel **170** may include a proximal mouth **702** and a distal stem **704**. The mouth **702** and the stem **704** may have circular cross sections. The mouth **702** may be conical or otherwise tapered. The diameter of the stem **704** may be much less than the diameter of the mouth **702**. The graft funnel **170** may include a central longitudinal hole **706** that extends through the graft funnel between the proximal end **698** and the distal end **700**. The hole **706** may include, from proximal to distal, first, second, third, and fourth inner portions **708**, **710**, **712**, **714**. Each of these inner portions may have a circular cross section. The first inner portion **708** may be conical or otherwise tapered, and may be shaped like the mouth **702**. The second inner portion **710** may be

described as a fillet radius transition or throat between the first inner portion **708** and the third inner portion **712**. The second inner portion **710** may be at the same or similar level as the transition between the mouth **702** and the stem **704**. The diameter of the third inner portion **712** may be less than the minor diameter of the first inner portion **708**. The fourth inner portion **714** may extend proximally about half or more of the proximal-distal length of the stem **704**. The diameter of the fourth inner portion **714** may be greater than the diameter of the third inner portion **712**. The proximal end of the fourth inner portion **714** may be flat, and a circumferential chamfer may be present at the inner edge between the third and fourth inner portions **712**, **714**.

(214) The stem **704** may be received by the second inner portion **256** of the hole **252** of the shuttle **122** of the shuttle assembly **114**. The fourth inner portion **714** may receive the proximal end **596** of the proximal lock tube end part **148** of the lock tube assembly **118**. The diameter of the third inner portion **712** may be the same as or similar to the diameter of the hole **606**.

(215) Referring to FIGS. **27**, **35**, and **36**, the tamp assembly **180** may include a tamp shaft **158**, a tamp shaft top part **168**, a tamp handle **172**, and a tamp shaft tip part **174**.

(216) The tamp shaft **158** extends between a proximal end **716** and a distal end **718**. The tamp shaft **158** may be a cylindrical shaft as shown. The tamp shaft **158** may include a proximal externally threaded portion **720** and a distal smooth portion **722**. The diameter of the distal smooth portion **722** may be greater than the major diameter of the proximal externally threaded portion **720**. The tamp shaft **158** may include a central longitudinal internally threaded hole **724** that extends proximally into the distal end **718**.

(217) The tamp shaft top part **168** extends between a proximal end **726** and a distal end **728**. The tamp shaft top part **168** may include a proximal externally threaded portion **730**, a middle smooth cylindrical portion **732**, and a distal torque fitting **734**, such as the hexagonal key shown. The diameter of the middle smooth cylindrical portion **732** may be greater than the major diameter of the proximal externally threaded portion **730**. The outer diameter of the distal torque fitting **734** may be less than the diameter of the middle smooth cylindrical portion **732**. The tamp shaft top part **168** may include a central longitudinal internally threaded hole **736** that extends proximally into the distal end **728**.

(218) The tamp handle **172** extends between a proximal end **738** and a distal end **740**. The tamp handle **172** may be a cylindrical part with rounded proximal and distal ends **738**, **740** as shown. The tamp handle **172** may include one or more friction features **742**, such as the series of shallow circumferential grooves shown. The tamp handle **172** may include a central longitudinal hole **744** that extends proximally into the distal end **740**. The hole **744** may extend proximally about half, or most, of the proximal-distal overall length of the tamp handle **172**. The hole **744** may include, from proximal to distal, first, second, and third inner portions **746**, **748**, **750** of approximately equal proximal-distal lengths. Each of the inner portions may have a circular cross section. The first inner portion **746** is not visible in FIG. **36**. The first inner portion **746** may be a blind hole included for ease of manufacturing the second inner portion **748**, which has internal threads. The minor diameter of the threads may be equal to or similar to the diameter of the first inner portion **750**. The diameter of the third inner portion **750** may be greater than the major diameter of the threads. The third inner portion **750** may be referred to as a counterbore around the second inner portion **748**. The proximal end of the third inner portion **750** may be flat.

(219) The tamp shaft tip part **174** extends between a proximal end **752** and a distal end **754**. The tamp shaft tip part **174** may be a cylindrical shaft as shown. The tamp shaft tip part **174** may include a proximal externally threaded portion **756** and a distal smooth portion **758**. The diameter of the distal smooth portion **758** may be greater than the major diameter of the proximal externally threaded portion **756**. The distal end **754** may have a concave surface **760**.

(220) The tamp assembly **180** may be assembled by threading the proximal externally threaded portion **730** of the tamp shaft top part **168** into the second inner portion **748** of the hole **744** of the tamp handle **172** so that the middle smooth cylindrical portion **732** is received in the third inner

portion **750**, threading the proximal externally threaded portion **720** of the tamp shaft **158** into the hole **736** of the tamp shaft top part **168**, and threading the proximal externally threaded portion **756** of the tamp shaft tip part **174** into the hole **724** of the tamp shaft **158**. The tamp assembly **180** may be a weldment. Alternatively, the tamp shaft **158**, tamp shaft top part **168**, tamp handle **172**, and/or tamp shaft tip part **174** may be combined into a single part.

(221) The outer diameters of the distal smooth portion **722** of the tamp shaft **158** and the distal smooth portion **758** of the tamp shaft tip part **174** may be received in the inner diameter of the third inner portion **712** of the hole **706** of the graft funnel **170** and in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

(222) Referring to FIGS. **29**, **30**, **37**, and **38**, the screwdriver assembly **190** may include a handle **176**, a shaft **178**, a screw holder **200**, and a screw holder catch **182**.

(223) The handle **176** extends between a proximal end **762** and a distal end **764**. The handle **176** may include a proximal grip portion **766** with one or more friction features **768** such as the four generally pear-shaped indentations shown, and a distal shaft **770**. The shaft **770** may include a proximal first portion **772** and a distal second portion **774**. The first portion **772** may include one or more friction features **776**, such as the series of shallow circumferential grooves shown. The first portion **772** may be a sleeve that is longitudinally movable relative to the second portion **774**. A central longitudinal D-shaped hole **778** may extend proximally into the distal end **764**. One or more balls (not shown) may be captured between the first and second portions **772**, **774** and may protrude at least partially into the D-shaped hole **778** when the first portion **772** is in a first position relative to the second portion **774**. The balls may be free to move radially away from the hole **778** when the first portion **772** is in a second position relative to the second portion **774**. The first portion **772** may be biased towards the first position.

(224) The shaft **178** extends between a proximal end **780** and a distal end **782**. The shaft **178** may be generally cylindrical as shown. The shaft **178** may include, from proximal to distal, first, second, third, fourth, and fifth outer portions **784**, **786**, **788**, **790**, **792**. The first outer portion **784** may have a generally circular cross section, and may include a unilateral longitudinal flat **794** so that the first outer portion **784** is complementary to the D-shaped hole **778** in the shaft **770** of the handle **176**. A circumferential groove **796** may extend around the circular side of the first outer portion **784** to engage the balls in the shaft **770** of the handle **176**. The diameter of the second outer portion **786** may be greater than the diameter of the first outer portion **784**. The second outer portion **786** may extend over most of the proximal-distal length of the shaft **178**. The third outer portion **788** may include external threads. The major diameter of the threads may be less than the diameter of the second outer portion **786**. The fourth outer portion **790** may have a circular cross section and may be smooth. The diameter of the fourth outer portion **790** may be the same as, or similar to, the minor diameter of the threads of the third outer portion **788**. The fifth outer portion **792** may include a torque fitting **798** such as the hexagonal key shown.

(225) Referring to FIGS. **39-42**, the screw holder **200** extends between a proximal end **800** and a distal end **802**. The screw holder **200** may be a tubular part with generally circular outer and inner cross-sectional shapes. The screw holder **200** may include, from proximal to distal, first and second outer portions **804**, **806**. The first outer portion **804** may have a generally circular cross section, and may extend distally over most of the proximal-distal length of the screw holder **200**. Bilateral flats **808**, **810** may be included at or near the middle of the proximal-distal length of the first outer portion. A longitudinal slot **812** may extend through one wall of the first outer portion **804**, and may be located between the flats **808**, **810**. The proximal end of the slot **812** may be at the same or similar level as the proximal ends of the flats **808**, **810**. The proximal end of the slot **812** may include a circular counterbore **814**. The distal end of the slot **812** may extend distally about twice as far as the distal ends of the flats **808**, **810**. The slot **812** may be rectangular or oval. The second

outer portion **806** may taper inwardly toward the distal end **802**. The second outer portion **806** may be conical. The second outer portion **806** may include a notch **816** that extends proximally from the distal end **802** through one wall of the second outer portion **806**. The notch **816** may be rectangular or square, and may have rounded corners. The notch **816** may be in line with the slot **812**. The screw holder **200** may include a central longitudinal hole **818** that extends through the screw holder between the proximal end **800** and the distal end **802**. The hole **818** may include, from proximal to distal, first, second, and third inner portions **820**, **822**, **824**. The first inner portion **820** may include internal threads. The diameter of the second inner portion **822** may be less than, equal to, or similar to the minor diameter of the threads. The proximal end of the second inner portion **822** may be proximal to the proximal ends of the flats **808**, **810**. The distal end of the second inner portion **822** may be distal to the proximal end of the notch **816**, and may be at or near the middle of the proximal-distal length of the notch **816**. A short tapered surface may be included at the transition between the second and third inner portions **822**, **824**. The second inner portion **822** may include a unilateral longitudinal alcove **826** extending into one wall of the second inner portion **822**. The alcove **826** may be in line with the slot **812** and the notch **816**. The diameter of the third inner portion **824** may be less than the major diameter of the threads of the first inner portion **820**, and may be greater than the minor diameter of the threads.

(226) The screw holder catch **182** extends between a proximal end **828** and a distal end **830**. The screw holder catch **182** may be a generally plate-like part, and may include, from proximal to distal, first, second, and third portions **832**, **834**, **836**. The first portion **832** may have a size and shape that is complementary to the slot **812** of the screw holder **200**. The proximal end of the first portion **832** may include a circular shelf **838** that is complementary to the counterbore **814**. The second portion **834** may be offset from the first portion **832** in a direction opposite the shelf **838**, and may be substantially parallel to the first portion **832**. The second portion **834** may include a longitudinal arcuate groove **840** that faces the same direction as the shelf **838**. The third portion **836** may have a size and shape that is complementary to the notch **816** of the screw holder **200**. The third portion **836** may be in line with the first portion **832** (thus offset from the second portion in a direction toward the shelf **838**). The third portion **836** may include a distal tooth **842** that faces the same direction as the shelf **838**.

(227) The screwdriver assembly **190** may be assembled by coupling the screw holder catch **182** to the screw holder **200** so that the shelf **838** is in the counterbore **814**, the second portion **834** is in the alcove **826**, the third portion **836** is in the notch **816**, and the groove **840** faces the interior of the screw holder **200**; threading the first inner portion **820** of the hole **818** of the screw holder **200** onto the third outer portion **788** of the shaft **178**; and inserting the first outer portion **784** of the shaft **178** into the D-shaped hole **778** of the handle **176** so that the balls engage the groove **796**. Alternatively, the handle **176**, shaft **178**, screw holder **200**, and/or screw holder catch **182** may be combined into a single part.

(228) The outer diameters of the first outer portion **804** of the screw holder **200** and the second outer portion **786** of the shaft **178** may be received in the inner diameter of the third inner portion **712** of the hole **706** of the graft funnel **170** and in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

(229) Referring to FIGS. **43** and **44**, the lockout screw **654** extends between a proximal end **844** and a distal end **846**. The lockout screw **654** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth portions **848**, **850**, **852**, **854**, **856**, **858**, each with a circular cross section. The first portion **848** may include a rounded outer proximal circumferential edge, and may extend a short distance in the proximal-distal direction. The first portion **848** may form a circumferential lip or flange around the proximal end of the lockout screw **654**. The diameter of the second portion **850** may be less than the diameter of the first portion **848**. There may be a 90 degree

corner between the first portion **848** and the second portion **850**. The diameter of the third portion **852** may be greater than the diameter of the first portion **848**. There may be a 90 degree corner between the second portion **850** and the third portion **852**, so that the second portion **850** forms a circumferential groove between the first and third portions **848**, **852**. The third portion **852** may have a rounded outer proximal circumferential edge. The diameter of the fourth portion **854** may be the same as or similar to the diameter of the first portion **848**, greater than the diameter of the second portion **850**, and less than the diameter of the third portion **852**. There may be a shoulder **655** between the third and fourth portions **852**, **854**. The shoulder **655** may taper, and may be conical. The fourth portion **854** may extend distally to about the middle of the proximal-distal length of the lockout screw **654**. The fifth portion **856** may include an externally threaded portion **653**. The major diameter of the threads may be the same as or similar to the diameter of the fourth portion **854**. There may be a tapered transition between the fourth and fifth portions **854**, **856**. The diameter of the sixth portion **858** may be the same as or similar to the minor diameter of the threaded portion **653**. The sixth portion **858** may have a smooth shaft and may include a rounded outer distal circumferential edge. The lockout screw **654** may include a torque fitting **860** in or on the proximal end **844**, such as the hexagonal socket shown. The lockout screw **654** is described in pending U. S. patent application Ser. No. 15/244,446.

(230) The outer diameter of the third portion **852** of the lockout screw **654** may be received in the inner diameter of the third inner portion **712** of the hole **706** of the graft funnel **170** and in the lock tube assembly **118**, more specifically in the hole **606** of the proximal lock tube end part **148**, the hole **594** of the proximal lock tube **124**, the second inner portion **646** of the hole **642** of the lock tube coupling **128**, and the hole **626** of the distal lock tube **126**.

(231) The lockout screw **654** may be coupled to the screwdriver assembly **190** by coupling the torque fitting **798** of the shaft **178** to the torque fitting **860** of the lockout screw **654**, inserting the first portion **848** of the lockout screw **654** into the third inner portion **824** of the screw holder **200**, and engaging the tooth **842** of the screw holder catch **182** in the second portion **850** of the lockout screw **654**. Alternatively, the lockout screw **654** may be coupled to the screwdriver assembly **190** by inserting the first portion **848** of the lockout screw **654** into the third inner portion **824** of the screw holder **200**, engaging the tooth **842** of the screw holder catch **182** in the second portion **850** of the lockout screw **654**, coupling the torque fitting **798** of the shaft **178** to the torque fitting **860** of the lockout screw **654**, and threading the first inner portion **820** of the hole **818** of the screw holder **200** onto the third outer portion **788** of the shaft **178**. The lockout screw **654** may be decoupled from the screwdriver assembly **190** by reversing these steps.

(232) Referring to FIGS. **45-47**, implant **1000** is expandable along a horizontal or lateral or transverse or medial-lateral axis **104** and a vertical or cephalocaudal axis **106**. Implant **1000** includes a first support member **1040** and a second support member **1030**, the support members connected by a first end body **1050** and a second end body **1052**. First support member **1040** may include an upper support body **1042** and a lower support body **1044**. A pair of connection pins **1055** project from the second end body **1052**. A plurality of link members **1060**, **1062**, **1064**, **1066** pivotably connect each of the first and second end bodies **1050**, **1052** to each of the support members **1030**, **1040**. Lower support body **1044** includes ramped expansion slots **1114**, **1124**, and upper support body includes ramped expansion slots **1154**, **1164**. The implant **1000** is representative of the intervertebral cages described in pending U.S. patent application Ser. No. 15/244,446.

(233) Referring to FIGS. **48-59**, methods for inserting and expanding an expandable intervertebral implant with instrument system **115** are shown. The instrument **100**, as seen in FIGS. **48** and **49**, includes a bayonet receiver **150** at the distal end **107** of the distal lock tube **126**. The bayonet receiver **150** includes a pair of shaped slots **152** which each include an approximately 90° bend. The slots are in the distal end **614** of the distal lock tube **126** of the lock tube assembly **118**. The bayonet receiver **150** may also include the inserter connector **138** of the shaft assembly **116**. In one

step of the method, the implant **1000** is attached to the distal end **107** of the instrument **100**. The connection pins **1055** are received in the shaped slots **152** of the bayonet receiver **150** so that the implant **1000** extends distally from the instrument **100**. The distal lock tube **126** is rotated so that the pins **1055** traverse the 90° bend of the shaped slots **152**, forming a bayonet connection **154** and locking the implant **1000** to the instrument **100**. The implant second end body **1052** may be stabilized against rotation by its fit within the third inner portion **550** of the inserter connector **138**. Other embodiments within the scope of the disclosure may include a threaded connection between the implant **1000** and instrument **100** instead of the bayonet connection. In an embodiment, the distal lock tube **126** may be rotated via rotation of the selector ring **144**. In an embodiment, the lock tube is rotated 30° to complete the bayonet connection. The lock tube assembly (distal lock tube **126**, lock tube coupling **128**, proximal lock tube **124**) may additionally be unlocked from the inserter shaft **136** and removed from the instrument **100** for cleaning.

(234) Referring to FIGS. **26**, **50**, and **51**, another step of the method includes inserting the draw bar assembly **160** distally through the instrument bore **112**. The distal threaded tip **162** of the draw bar assembly **160** passes through the implant second end body **1052** and is threadably engaged with the implant first end body **1050**. The instrument lock collar **120** may be actuated to engage the instrument shuttle **122** with an annulus **166** near the proximal end of the draw bar assembly **160**. The lock collar **120** may be moved to its first position, the draw bar assembly **160** may be advanced within the instrument bore **112** until the annulus is distal to the lock collar **120**, and the lock collar **120** may be moved to its second position. When the draw bar assembly **160** is inserted into connection with implant **1000** but before expansion of the implant, indicia **248** or markings may indicate the implant is in the collapsed configuration.

(235) Referring to FIGS. **52** and **53**, the instrument **100** may be used to insert the implant **1000** into a prepared intervertebral space between a first vertebra **2** and a second vertebra (not shown for clarity), in another step of the method. The implant **1000** may be positioned in the intervertebral space before or after insertion of the draw bar assembly **160** into the instrument **100**.

(236) When the implant is properly positioned in the intervertebral space and the draw bar assembly **160** is inserted into the instrument **100**, in another step of the method the expansion knob **140** is rotated, turning threaded driver **130**. The knob **140**, threaded driver **130** and shuttle **122** form a motion conversion mechanism to convert rotational motion to linear or axial motion along the longitudinal instrument axis **102**. Interaction of threaded driver **130** with shuttle **122** translates shuttle **122** and draw bar assembly **160** proximally, providing axial force in a first direction along axis **102** to draw the implant first end **1050** proximally toward the implant second end **1052** to laterally expand the implant in a second direction perpendicular to the first direction, along axis **104**. As the axial force is applied to the implant, the link members **1060**, **1062**, **1064**, **1066** pivot outward away from the central longitudinal axis **102**, carrying the first and second support members **1040**, **1030** away from one another along axis **104**, laterally expanding the implant. Stop surfaces on the implant end bodies and the support members may interact to limit the lateral expansion. When the lateral expansion is attained, indicia **248** or markings on the instrument may indicate the implant is in the laterally expanded configuration.

(237) In another step of the method, the implant **1000** is expanded vertically. Referring to FIGS. **54** and **55**, the expansion knob **140** is rotated farther to continue to translate draw bar assembly **160** proximally along the longitudinal instrument axis **102**. The continued axial force along the first direction causes the link members **1064**, **1066** within the implant **1000** to travel along the expansion slots **1114**, **1124**, **1154**, **1164** in the upper and lower support bodies **1042**, **1044**, urging the upper and lower support bodies away from one another and vertically expanding the first support member **1040** along a third direction indicated by axis **106**, perpendicular to the first and second directions. When the vertical expansion is attained, indicia **248** or markings on the instrument may indicate the implant is in the vertically expanded configuration.

(238) It is appreciated that the expansion knob may be actuated in discrete steps to first laterally

and then vertically expand the implant, or the expansion knob may be actuated in a continuous motion to first laterally and then vertically expand the implant. It is also appreciated that in another embodiment, the vertical expansion of the implant may precede the horizontal expansion. In the embodiment shown, only the first support member **1040** expands vertically; the height of the second support member **1030** is fixed. This may provide a lordotic correction between the first and second vertebrae. In other embodiments, the first support member **1040** may be fixed and the second support member **1030** may be vertically expandable, or the first and second support members **1040**, **1030** may both be vertically expandable. When expansion is complete, the draw bar assembly **160** may be disengaged from the shuttle **122** by actuating the lock collar **120**, unthreaded from the implant **1000** and removed from the instrument **100**.

(239) In another step of the method, graft material may be inserted through the instrument into the expanded implant **1000**. Referring to FIGS. **27**, **28**, **56**, and **57**, funnel **170** may be connected to the proximal end of instrument bore **112**. Graft material may be inserted axially through the funnel **170** and bore **112** and deposited into the implant inner chamber **1120**. The tamp assembly **180** may be deployed through instrument **100** to push and pack the graft through the bore **112** and in the chamber **1120**.

(240) In another step of the method, a locking fastener such as lockout screw **654** may be inserted through the instrument and into fixed engagement with the expanded implant **1000**, to lock the implant in the horizontally and vertically expanded configuration. Referring to FIGS. **29**, **30**, **58**, and **59**, screw **654** may be loaded into screw holder **200** and connected to a distal end of screwdriver assembly **190**. A distal tip of screwdriver assembly **190** may include a tip **792** shaped to complementarily engage the proximal end of screw **654**. Screwdriver assembly **190**, with attached screw holder **200** and screw **654**, may be inserted longitudinally through bore **112** of instrument **100**. The screwdriver assembly **190** may be actuated to engage a threaded portion **653** of screw **654** with the threaded first end body **1050** of implant **1000**. As the screw is driven distally, a proximal shoulder **655** of screw **654** may abut a surface of the second end body **1052**, rigidly locking the cage in the expanded configuration. Screw holder **200** may include a release feature or mechanism to release screw **654** from the holder **200** after the screw is in engagement with implant **1000**. For example, the screw holder catch **182** may be biased to disengage the tooth **842** from the second portion **850** of the screw **654** when the screw is engaged with the implant **1000**. When deployment and release of screw **654** is complete, the screwdriver assembly **190** and attached screw holder **200** may be withdrawn from the instrument **100**.

(241) Referring to FIGS. **60-63**, implant **1000** is shown in situ after insertion between the first vertebral body **2** and the second vertebral body. In the embodiment depicted, support members **1040** and **1030** have been expanded horizontally along an anterior-posterior direction to provide support between the first and second vertebral bodies. Support member **1040** has been expanded vertically along a cephalocaudal direction to provide a lordotic correction between the first and second vertebrae. It is appreciated that in other embodiments of the invention, both support members could be expanded vertically, or support member **1030** could be solely expanded vertically to provide a kyphotic correction. One or more implants **1000** may be implanted between first and second vertebrae. Other approaches may be used within the scope of the disclosure, including lateral, transforaminal and anterior. One or more implants may be inserted in any orientation relative to the first and second vertebrae.

(242) Referring to FIGS. **64-71**, another insertion and expansion instrument is shown. Instrument **300** may be deployed to insert and expand implant **1000** or another expandable intervertebral implant. Instrument **300** may include a bayonet or other connection to removably connect implant **1000** to the instrument. Instrument **300** includes a ratchet mechanism **310** operable by a lever **312** to axially translate a shuttle **322** and draw bar assembly **328** proximally. The ratchet mechanism **310** may be referred to as an actuator. Similar to instrument **100**, translation of draw bar assembly **328** provides axial force to laterally and/or vertically expand the connected implant. Other tools

such as the funnel **170**, tamp assembly **180**, and screwdriver assembly **190** disclosed herein may be used, or adapted for use, with instrument **300** to deposit graft material within the implant and secure a locking fastener such as screw **654**.

(243) Instrument **300** extends between a proximal end **862** and a distal end **864** along a longitudinal instrument axis **866**. Instrument **300** may include the ratchet mechanism **310**, the lever **312**, the shuttle **322** which may be similar to shuttle **122**, an inserter handle **302** which may be similar to inserter handle **142**, a shaft assembly **304** which may be similar to shaft assembly **116**, an inserter shaft **306** which may be similar to inserter shaft **136**, a distal tip part **308** which may be similar to inserter connector **138**, a lock lever **314**, a stop screw **316**, a lock tube assembly **318** which may be similar to lock tube assembly **118**, a lock tube coupling **320** which may be similar to lock tube coupling **128**, a proximal lock tube **324** which may be similar to proximal lock tube **124**, a distal lock tube **326** which may be similar to distal lock tube **126**, a draw bar assembly **328** which may be similar to draw bar assembly **160**, a draw bar knob **330** which may be similar to handle **164**, a draw bar tip part **332**, a draw bar **334** which may be similar to draw bar **156**, a spring plunger **336**, a ratchet **338**, a lever link **340**, a return link **342**, and/or a release ring **344**.

(244) Referring to FIGS. **70-73**, the ratchet mechanism **310** may include the lever **312**, spring plunger **336**, ratchet **338**, lever link **340**, and return link **342**. The ratchet mechanism **310** may include the shuttle **322**.

(245) The lever **312** extends between a proximal end **868** and a distal end **870**. The lever **312** may include a body **872** and an arm **874**. The arm **874** may extend proximally obliquely away from an outer side **876** of the body **872**. An opposite inner side **878** of the body may include a pair of outer arms **880**, **882** forming an outer clevis, and a pair of inner arms **884**, **886** forming an inner clevis. The inner arms **884**, **886** may be located between the outer arms **880**, **882**. The inner arms **884**, **886** may be shorter than the outer arms **880**, **882**, and may extend from the inner side **878** about one third as far as the outer arms. The arms **880**, **882**, **884**, **886** may all be parallel as shown, and may be perpendicular to the inner side **878**. A first hole **888** may extend through the arms **880**, **882**, **884**, **886** near the inner side **878**. A second hole **890** may extend through the outer arms **880**, **882** near the free ends of the outer arms. The first and second holes **888**, **890** may be parallel to the inner side **878**. A third hole **892** may extend into the inner side **878** between the inner arms **884**, **886**. The third hole **892** may be parallel to the arms **880**, **882**, **884**, **886**.

(246) The spring plunger **336** extends between a proximal end **892** and a distal end **894**. The spring plunger **336** may be a generally cylindrical shaft with circular cross sections as shown. The spring plunger **336** may include, from proximal to distal, first and second outer portions **896**, **898**. The first outer portion **896** may extend distally most of the proximal-distal length of the spring plunger **336**. The diameter of the second outer portion **898** may be less than the diameter of the first outer portion **896**. There may be a tapered transition between the first and second outer portions **896**, **898**. The second outer portion **898** may include a rounded distal outer edge. A central longitudinal hole **900** may extend distally into the proximal end of the spring plunger **336**. The hole **900** may extend distally almost as far as the first outer portion **896**.

(247) The ratchet **338** extends between a proximal end **902** and a distal end **904**. The ratchet **338** may have a rectangular or square cross section as shown. A tooth **906** may protrude from an inner side **908** of the ratchet **338**. The tooth **906** may be located along an inner proximal edge of the ratchet **338**. A first hole **910** may extend through the ratchet **338** near the distal end **904** and may be parallel to the tooth **906**. A second hole **912** may extend into the inner side **908** near the distal end **904** and may be perpendicular to the first hole **910**. The second hole **912** may be between the first hole **910** and the distal end **904**.

(248) The lever link **340** extends between a proximal end **914** and a distal end **916**. The lever link **340** may have a rectangular or square cross section as shown. A tooth **918** may protrude from an inner side **920** of the lever link **340**. The tooth **918** may be located along an inner proximal edge of the lever link **340**. A unilateral notch **922** may extend along the lever link **340** so as to intersect a

proximal portion of the inner side **920** and an inner portion of the proximal end **916**, so that the tooth **918** is thinner than the distal end **916**. A first hole **924** may extend through the lever link **340** near the distal end **916** and may be parallel to the tooth **918**. A second hole **926** may extend into an outer side **928** of the lever link **340** opposite the inner side **920** near the distal end **916** and may be perpendicular to the first hole **924**. The first hole **924** may be between the second hole **926** and the distal end **916**.

(249) The return link **342** extends between a proximal end **930** and a distal end **932**. The return link **342** may have a rectangular or square cross section as shown. A foot **934** may protrude from a first side **936** of the return link **342**. The foot **934** may be located at the proximal end **930** of the return link **342**. The foot **934** may extend perpendicular to the proximal-distal length of the return link **342**. A unilateral notch **938** may extend along the return link **342** so as to intersect a proximal portion of an inner side **940** of the return link and an inner portion of the proximal end **930**. A first hole **942** may extend distally into the proximal end **930** and the foot **934**. A second hole **944** may extend through the return link **342** near the distal end **932** and may be parallel to the foot **934**.

(250) The shuttle **322** extends between a proximal end **946** and a distal end **948**. The shuttle **322** may include, from proximal to distal, first, second, third, and fourth outer portions **950**, **952**, **954**, **956**. The first outer portion **950** may have a circular cross section and rounded proximal and distal outer circumferential edges. The proximal-distal length of the first outer portion **950** may be less than its diameter. The second outer portion **952** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **950**. The second outer portion **952** may form a square (90 degree) inner corner with the first outer portion **950**. The proximal-distal length of the second outer portion **952** may be the same as or similar to the length of the first outer portion **950**.

(251) The third outer portion **954** may have a circular cross section with a diameter that is less than the diameter of the second outer portion **952**. The third outer portion **954** may form a square inner corner with the second outer portion **952**. The third outer portion **954** may extend over most of the proximal-distal length of the shuttle **322**. The fourth outer portion **956** may have a circular cross section that tapers down in diameter toward the distal end **948**. The fourth outer portion **956** may be conical. A unilateral longitudinal groove **958** may extend along the second and third outer portions **952**, **954** and may run out in the fourth outer portion **956**. The groove **958** may have a rectangular or square cross section with a rounded proximal end at the proximal end of the second outer portion **952**. A first series of transverse grooves **960** may extend across the third outer portion **954** and may be located 90 degrees from the longitudinal groove **958**. There may be ten grooves **960** as shown, or another number. The size and shape of the grooves **960** may be complementary to the tooth **918** of the lever link **340**. A second series of transverse grooves **962** may extend across the third outer portion **954** and may be located 90 degrees from the longitudinal groove **958** and 180 degrees from the first series of transverse grooves **960**. There may be 54 grooves **962** as shown, or another number. The size and shape of the grooves **960** may be complementary to the tooth **906** of the ratchet **338**. Indicia **964** may be included along the third outer portion **954** opposite the longitudinal groove **958**. The indicia **964** may include transverse lines and numerals as shown, or other markings or engravings. A unilateral shaped slot **966** may extend proximally into the distal end of the third outer portion **954** and may run out in the fourth outer portion **956**. The proximal end of the slot **966** may make a 90 degree bend toward the longitudinal groove **958**. The slot **966** may have a rectangular or square cross section. The slot **966** and other similarly shaped slots disclosed herein may be referred to as a bayonet slot.

(252) The shuttle **322** may include a central longitudinal hole **968** that extends through the shuttle between the proximal end **946** and the distal end **948**. The hole **968** may include, from proximal to distal, first, second, third, and fourth inner portions **970**, **972**, **974**, **976**, each with a circular cross section. The first inner portion **970** may be a shallow countersink around the proximal end of the hole **968**. The second inner portion **972** may have internal threads. The minor diameter of the threads may be the same as or similar to the minor diameter of the first inner portion **970**. The third

inner portion **974** may be smooth, with a diameter that is less than the minor diameter of the threads of the second inner portion **972**. See FIG. **66**. The distal end of the third inner portion **974** may have a square (90 degree) inner corner. The fourth inner portion **976** may be smooth, with a diameter that is less than the diameter of the third inner portion **974**.

(253) Referring to FIGS. **70**, **71**, **74**, and **75**, the shaft assembly **304** may include the inserter shaft **306** and distal tip part **308**.

(254) The inserter shaft **306** extends between a proximal end **978** and a distal end **980**. The inserter shaft **306** may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft **306** may include a pair of shaped slots **982**, **984** which each include an approximately 90 degree bend. The slots **982**, **984** may extend longitudinally into the proximal end **978** before making the bend. Each slot **982**, **984** may extend through one wall of the inserter shaft **306**. The inserter shaft **306** may include bilateral transverse grooves **986**, **988** which may be located 90 degrees from, and distal to, the slots **982**, **984**. The inserter shaft **306** may include one or more openings **990**. Three openings **990** are shown spaced along the inserter shaft **306** distal to the transverse grooves **986**, **988**. Each opening **990** may extend through one wall of the inserter shaft **306**, or through both walls as shown. The inserter shaft **306** may include a central longitudinal hole **992** that extends through the inserter shaft between the proximal end **978** and the distal end **980**. The hole **992** may include a proximal countersink **994** and/or a distal counterbore **996**.

(255) The distal tip part **308** extends between a proximal end **1200** and a distal end **1202**. The proximal end **1200** may be circular in a proximal view. The distal end **1202** may be rectangular in a distal view. The distal end **1202** may have rounded corners in the distal view. The distal tip part **308** may include, from proximal to distal, first, second, and third outer portions **1204**, **1206**, **1208**. The first outer portion **1204** may have a circular cross section and may extend distally about one fourth of the proximal-distal length of the distal tip part **308**. The second outer portion **1206** may have a circular cross section and may have a diameter that is greater than the diameter of the first outer portion **1204**. The second outer portion **1206** may form a square (90 degree) inner corner with the first outer portion **1204**. The third outer portion **1208** may include the rectangular distal end **1202**, and may also include a curvaceous transition surface that blends with the second outer portion **1206**. The distal tip part **308** may include a central longitudinal hole **1210** that extends through the distal tip part between the proximal end **1200** and the distal end **1202**. The hole **1210** may include, from proximal to distal, first and second inner portions **1212**, **1214**. The first inner portion **1212** may have a circular cross section and may extend distally about two thirds to three fourths of the overall proximal-distal length of the distal tip part **308**. The cross-sectional shape of the second inner portion **1214** may be rectangular or square. The rectangle or square may have rounded corners. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion **1212**, and may be aligned with the major dimension (length) of the rectangle of the distal end **1202**. The minor dimension (width) of the rectangle may be greater than the diameter of the first inner portion **1212**. The second inner portion **1214** may have a flat proximal end.

(256) The shaft assembly **304** may be assembled by inserting the first outer portion **1204** of the distal tip part **308** into the counterbore **996** of the inserter shaft **306** and aligning the length of the rectangle of the third outer portion **1208** to extend over the transverse grooves **986**, **988**. The inserter shaft **306** and distal tip part **308** may be permanently connected, for example by welding. Alternatively, the inserter shaft **306** and distal tip part **308** may be combined into a single part. The inner diameters of the hole **992** of the inserter shaft **306** and the first inner portion **1212** of the hole **1210** of the distal tip part **308** may be the same or similar, and together they may form a shaft assembly bore **1434**. See FIGS. **70** and **71**.

(257) Referring to FIGS. **70**, **71**, **76**, and **77**, the lock tube assembly **318** may include the lock tube coupling **320**, proximal lock tube **324**, and distal lock tube **326**.

(258) The lock tube coupling **320** extends between a proximal end **1216** and a distal end **1218**. The lock tube coupling **320** may be a tubular part. The lock tube coupling **320** may include a proximal

body **1220** which may have a circular cross-sectional outer shape, and may extend distally about half or three-fifth of the overall proximal-distal length of the lock tube coupling **320**. A first lateral post **1222** may extend radially outwardly from the distal end of the body **1220**. A unilateral tab **1224** may protrude distally from the body **1220**, and may be located in line with the first lateral post **1222**. The outer surface of the tab **1224** may be radially inset relative to the outer diameter of the body **1220**. A second lateral post **1226** may protrude radially outwardly from the tab **1224**. The second lateral post **1226** may protrude radially beyond the outer diameter of the body **1220** as shown. The second lateral post **1226** may be distal to and in line with the first lateral post **1222**. First and second holes **1228**, **1230** may extend through one wall of the body **1220** on either side of a location opposite the first lateral post **1222**, or in other words, each hole may be located about 135 degrees away from the first lateral post **1222**. The lock tube coupling **320** may include a central longitudinal hole **1232** that extends through the lock tube coupling between the proximal end **1216** and the distal end **1218**. The hole **1232** may include, from proximal to distal, first, second, and third inner portions **1234**, **1236**, **1238**. The first inner portion **1234** may have a circular cross section and may extend distally about a third of the overall proximal-distal length of the lock tube coupling **320**, or about half of the length of the body **1220**. The second inner portion **1236** may have a circular cross section with a diameter less than the diameter of the first inner portion **1234**. The second inner portion **1236** may extend distally only a short distance as a complete diameter, but a section of it may extend through the distal end **1218** to form the inner side of the tab **1224**. The third inner portion **1238** may have a non-circular cross section that is a C-shaped alcove in the distal wall of the body **1220** opposite the tab **1224**. The alcove may extend around more than 180 degrees of arc.

(259) The proximal lock tube **324** extends between a proximal end **1240** and a distal end **1242**. The proximal lock tube **324** may be a tubular part with circular outer and inner cross-sectional shapes and constant outer and inner diameters.

(260) The distal lock tube **326** extends between a proximal end **1244** and a distal end **1246**. The distal lock tube **326** may be a tubular part with circular outer and inner cross-sectional shapes. The proximal end **1244** may include a unilateral notch **1248** that extends distally into one side of the distal lock tube **326**. The notch **1248** may extend around less than 180 degrees of arc. The notch **1248** may include, from proximal to distal, first and second portions **1250**, **1252**. The first portion **1250** may have a flat distal end with square corners. The second portion **1252** may be narrower than the first portion **1250**. The second portion **1252** may have a flat distal end with round corners. The distal end **1246** may include a pair of shaped slots **1254**, **1256** which each include an approximately 90° bend. The slots **1254**, **1256** may extend longitudinally into the distal end **1246** before making the approximately 90° bend. Each slot **1254**, **1256** may extend through one wall of the distal lock tube **326**. The distal lock tube **326** may include a central longitudinal hole **1258** that extends through the distal lock tube between the proximal end **612** and the distal end **614**. The hole **1258** may include, from proximal to distal, first and second inner portions **1260**, **1262**. The first inner portion **1260** may extend distally for most of the proximal-distal length of the distal lock tube **326**, and may have a flat distal end. The diameter of the second inner portion **1262** may be less than the diameter of the first inner portion **1260**.

(261) The lock tube assembly **318** may be assembled by inserting the distal end **1242** of the proximal lock tube **324** in the first inner portion **1234** of the lock tube coupling **320**, and inserting the proximal end **1244** of the distal lock tube **326** in the third inner portion **1238** of the lock tube coupling **320** so that the tab **1224** is received in the second portion **1252** of the notch **1248**. The lock tube coupling **320**, proximal lock tube **324**, and distal lock tube **326** may be permanently connected, for example by welding. Alternatively, the lock tube coupling **320**, proximal lock tube **324**, and/or distal lock tube **326** may be combined into a single part. The inner diameters of the second inner portion **1236** of the hole **1232** of the lock tube coupling **320**, proximal lock tube **324**, and first inner portion **1260** of the hole **1258** of the distal lock tube **326** may all be the same or

similar, and together they may form a longitudinal instrument bore **1260**. See FIGS. **64**, **70**, and **71**. The distal lock tube **326** and the body **1220** of the lock tube coupling **320** may fit within the bore **1434** of the shaft assembly **304**. The post **1226** of the lock tube coupling **320** may fit within either slot **982**, **984** of the inserter shaft **306** of the shaft assembly **304**.

(262) Referring to FIGS. **70**, **71**, **78**, and **79**, the draw bar assembly **328** may include the draw bar knob **330**, draw bar tip part **332**, and draw bar **334**.

(263) The draw bar knob **330** extends between a proximal end **1262** and a distal end **1264**.

(264) The draw bar handle **330** extends between a proximal end **1262** and a distal end **1264**. The draw bar handle **330** may include, from proximal to distal, a knob body **1266** and a shaft **1268**. The knob body **1266** may be generally disc-shaped with an outer diameter that is greater than its proximal-distal length. The knob body **1266** may include one or more grip features **1270** such as the six rim scallops shown. The outer diameter of the shaft **1268** may be less than the outer diameter of the knob body **1266** and the proximal-distal length of the shaft may be greater than the proximal-distal length of the knob body **1266**. The shaft **1268** may include a flange **1272** or annulus that extends circumferentially around a middle portion of the shaft. The flange **1272** may have square proximal and distal outer circumferential edges. The shaft **1268** may include one or more holes **1274** that extend transversely into, or through, the shaft; two through holes **1274** are shown between the distal side of the knob body **1266** and the proximal side of the flange **1272**. The two through holes **1274** are perpendicular to each other. A distal portion of the shaft **1268** may include external threads **1276**. The outer diameter of the flange **1272** may be greater than the major diameter of the threads **1276**. The minor diameter of the threads **1276** may be greater than the outer diameter of the remaining smooth portions of the shaft **1268**. The draw bar handle **330** may include a central longitudinal hole **1278** that extends through the draw bar handle between the proximal end **1262** and the distal end **1264**. The hole **1278** may include, from proximal to distal, first, second, and third inner portions **1280**, **1282**, **1284**. The first inner portion **1280** may have a circular cross section and may extend distally about half, or most, of the proximal-distal length of the knob body **1266**. The first inner portion **1280** may have a flat distal end. A circumferential fillet radius may be present around the proximal inner edge of the first inner portion **1280**. A circumferential fillet radius may be present in the distal inner corner of the first inner portion **1280**. The second inner portion **1282** may have a circular cross section and may extend distally between the distal side of the knob body **1266** and the holes **1274**. The diameter of the second inner portion **1282** may be less than the diameter of the first inner portion **1280**. The third inner portion **1284** may have a circular cross section and may extend distally through the distal end of the **1264** of the draw bar handle **330**. The diameter of the third inner portion **1284** may be less than the diameter of the first inner portion **1280** and greater than the diameter of the second inner portion **1282**. The third inner portion **1284** may have a flat proximal end. The hole(s) **1274** may extend into a proximal end of the third inner portion **1284**.

(265) The draw bar tip part **332** extends between a proximal end **1286** and a distal end **1288**. The draw bar tip part **332** may include, from proximal to distal, first, second, third, and fourth outer portions **1290**, **1292**, **1294**, **1296**. The first outer portion **1290** may have a circular cross section and may extend distally about one third to half of the overall proximal-distal length of the draw bar tip part **332**. An outer proximal edge of the first outer portion **1290** may include a circumferential chamfer. The second outer portion **1292** may have a circular cross section and may extend distally only a short distance. The diameter of the second outer portion **1292** may be greater than the diameter of the first outer portion **1290**. The second outer portion **1292** may be referred to as a flange or annulus around a middle portion of the draw bar tip part **332**. The third outer portion **1294** may have a circular cross section. The diameter of the third outer portion **1294** may be greater than the diameter of the first outer portion **1290** and less than the diameter of the second outer portion **1292**. The fourth outer portion **1296** may be a torque fitting, such as the hexagonal key shown. An outer distal edge of the fourth outer portion **1296** may include a circumferential chamfer. The draw

bar tip part **332** may include a central longitudinal hole **1298** that extends through the draw bar tip part between the proximal end **1286** and the distal end **1288**. The hole **1298** may include, from proximal to distal, first, second, and third inner portions **1300**, **1302**, **1304**. The first inner portion **1300** may have a circular cross section. The distal end of the first inner portion **1300** may be proximal to the second outer portion **1292**. The second inner portion **1302** may include internal threads. The major diameter of the threads may be less than the diameter of the first inner portion **1300**. The third inner portion **1304** may have a circular cross section. The diameter of the third inner portion **1304** may be less than the diameter of the first inner portion **1300** and greater than the major diameter of the threads of the second inner portion **1302**.

(266) The draw bar **334** extends between a proximal end **1306** and a distal end **1308**. The draw bar **334** may be a cylindrical shaft as shown. The draw bar **334** may include a central longitudinal hole **1310** that extends proximally into the distal end **1308** of the draw bar **334**. The hole **1310** may include, from proximal to distal, first and second inner portions **1312**, **1314**. The first inner portion **1312** may have a circular cross section. The second inner portion **1314** may have a circular cross section with a diameter greater than the diameter of the first inner portion **1312**.

(267) The draw bar assembly **328** may be assembled by inserting the proximal end **1306** of the draw bar **334** in the second inner portion **1282** of the hole **1278** of the draw bar handle **330** so that the proximal end **1306** is at the same or similar level with the distal end of the first inner portion **1280**, and inserting the first outer portion **1290** of the draw bar tip part **332** in the second inner portion **1314** of the hole **1310** of the draw bar **334**. A circumferential gap **1316** may exist between the outer surface of the draw bar **334** and the inner surface of the third inner portion **1284**. See FIG. **66**. The draw bar knob **330**, draw bar tip part **332**, and draw bar **334** may be permanently connected, for example by welding. Alternatively, the draw bar knob **330**, draw bar tip part **332**, and/or draw bar **334** may be combined into a single part. The draw bar tip part **332** and draw bar **334** may fit within the bore **1260** of the lock tube assembly **318**. The proximal end **1240** of the proximal lock tube **324** may fit in the gap **1316**.

(268) Referring to FIGS. **66**, **70**, **71**, and **80-85**, the inserter handle **302** extends between a proximal end **1318** and a distal end **1320**. The inserter handle **302** may include, from proximal to distal, first, second, and third outer portions **1322**, **1324**, **1326**. The first outer portion **1322** may be cylindrical with a circumferential groove **1328** centrally located along the proximal-distal length of the first outer portion **1322**. The groove **1328** may have square inner corners. The second outer portion **1324** may extend distally over most of the overall proximal-distal length of the inserter handle **302**. The second outer portion **1324** may be generally hourglass shaped or otherwise contoured to be comfortably and ergonomically gripped by a user's hand. The third outer portion **1326** may extend over about one fourth of the overall proximal-distal length of the inserter handle **302**. The third outer portion **1326** may include flat front and back surfaces **1330**, **1332** which may extend proximally from the distal end **1320** over the third outer portion **1326** and most of the proximal-distal length of the second outer portion **1324**. The third outer portion **1326** may include a rectangular or square boss **1374** which protrudes from a right side **1376** of the third outer portion. The boss **1374** may extend from a middle portion of the proximal-distal length of the third outer portion **1326**. A left side **1378** of the third outer portion **1326** may be a portion of a cylinder.

(269) The inserter handle **302** may include a unilateral recess **1334** that extends distally into the second outer portion. The recess **1334** may be on the same side as the front surface **1330**. The recess **1334** may be arcuate, and may extend around less than 180 degrees of arc. The recess **1334** may include bilateral ridges **1336**, **1338** as seen best in FIG. **82**. The ridges **1336**, **1338** may extend proximally a short distance from the distal end of the recess **1334**. A pocket **1340** may be located in the proximal region of the front surface **1330**. A window **1342** may be located within the pocket **1340**. The window **1342** may extend through the front side of the inserter handle **302**. The pocket **1340** may include indicia **1344** such as the transverse lines shown bracketing the window **1342**. A hole **1380** may extend through the inserter handle **302** between the front and back sides and may be

located in the proximalright portion of the third outer portion **1326**.

(270) A longitudinal slot **1346** may extend into a right side **1348** of the second outer portion **1324** of the inserter handle **302**. The slot **1346** may be rectangular with rounded corners. The proximal end of the slot **1346** may be proximal to the pocket **1340** and the distal end of the slot may be distal to the pocket. A distal portion **1350** of the slot **1346** may be shallow, with a flat bottom, and a proximal portion **1352** of the slot may be deep. The distal portion **1350** may extend over the distal one third of the overall proximal-distal length of the slot **1346**. The distal portion **1350** may include a hole **1354** in the bottom near the distal end. A transverse hole **1356** may extend through the inserter handle **302** between the front and back sides so as to intersect the side walls of the distal portion **1350** proximal to the hole **1354**. A hole **1358** may extend into the right side **1348** of the second outer portion **1324** of the inserter handle **302** distal to the slot **1346**. The hole **1358** may include internal threads and a counterbore to complement the stop screw **316**. A transverse slot **1382** may extend into the right side **1376** of the third outer portion **1326**. The slot **1382** may extend into a distal portion of the boss **1374**. The slot **1382** may taper in the transverse direction, becoming narrower as the slot extends into the third outer portion **1326**. See FIG. **82**.

(271) A longitudinal slot **1360** may extend into a left side **1362** of the second and third outer portions **1324**, **1326** of the inserter handle **302**. The proximal end of the slot **1360** may be at the same or similar level as the proximal end of the slot **1346** and the distal end of the slot may extend about half of the overall proximal-distal length of the third outer portion **1326**. A proximal portion **1364** of the slot **1360** may be narrower than a distal portion **1366** of the slot. The proximal and distal portions **1364**, **1366** may each be rectangular with rounded corners. The distal end of the proximal portion **1364** may be distal to the hole **1358** and proximal to the third outer portion **1326**. A unilateral narrow ledge **1368** may extend along a proximal back side wall within the proximal portion **1364**. A platform **1370** may extend between the side walls of the proximal and distal portions **1364**, **1366**. The proximal end of the platform **1370** may be distal to the hole **1358** and proximal to the distal end of the proximal portion **1364**, and the distal end of the platform may be located about one third of the overall proximal-distal length of the third outer portion **1326**, or about one half of the overall proximal-distal length of the distal portion **1366**. The platform **1370** may be deeper within the inserter handle **302** than the ledge **1368**. A longitudinal hole **1372** may extend proximally into the distal end **1320** of the inserter handle **302** and into the proximal end wall of the proximal portion **1364** of the slot **1360**. See FIG. **66**.

(272) The inserter handle **302** may include a central longitudinal hole **1384** that extends through the inserter handle between the proximal end **1318** and the distal end **1320**. The hole **1384** may include, from proximal to distal, first, second, and third inner portions **1386**, **1388**, **1389**. Each of these inner portions may have a circular cross section. The first inner portion **1386** may extend distally to the proximal end of the platform **1370**. The diameter of the second inner portion **1388** may be less than the diameter of the first inner portion **1386**. The second inner portion **1388** may extend distally to intersect the slot **1382**. The diameter of the third inner portion **1389** may be less than the diameter of the second inner portion **1388**. The window **1342**, proximal portion **1352** of the slot **1346**, hole **1358**, and proximal portion **1364** of the slot **1360** may open into the first inner portion **1386**. The distal portion **1366** of the slot **1360**, distal to the platform **1370**, and the slot **1382** may open into the second inner portion **1388**. Bilateral holes **1390**, **1392** may extend through the inserter handle **302** between the right and left sides near the distal end **1320**. The holes **1390**, **1392** may intersect the third inner portion **1389**.

(273) The lock lever **314** extends between a proximal side **1394** and a distal side **1395**. The lock lever **314** may include a ring **1398**, a stem **1400**, and a cap **1402**. The stem **1400** may extend from one side of the ring **1398**. The cap **1402** may be an enlargement at the free end of the stem **1400**. A central longitudinal hole **1404** may extend through the lock lever **314** between the proximal and distal sides **1394**, **1396**. A keyway **1406** may also extend through the lock lever **314** between the proximal and distal sides **1394**, **1396** along an inner wall of the hole **1404** at the base of the stem

1400. A hole **1408** may extend into the stem **1400** near the ring **1398**.

(274) The stop screw **316** may be a dog point screw that is complementary to the hole **1358**. The stop screw **316** may have a pan head or cheese head.

(275) The release ring **344** extends between a proximal end **1410** and a distal end **1412**. The release ring **344** may include one or more friction features **1414**, such as the twelve rim scallops shown. The release ring **344** may include a unilateral tab **1416** that may extend distally. An inner side of the tab **1416** may include two longitudinal grooves **1418**, **1420** that extend through the distal end **1412**. A central longitudinal hole **1422** may extend through the release ring **344** between the proximal and distal sides **1410**, **1412**. The hole **1422** may include, from proximal to distal, first and second inner portions **1424**, **1426**. The first inner portion **1424** may have a generally circular cross section with a unilateral key **1428** extending radially into the hole **1422** at the proximal end **1410**. The key **1428** may be generally opposite the tab **1416**, but may be offset several degrees from being exactly opposite the tab. For example, the key **1428** may be 155 degrees away from the tab **1416**. The second inner portion **1426** may have a circular cross section with a diameter that is greater than the diameter of the first inner portion **1424**. The proximal end of the second inner portion **1426** may be flat. The inner side of the tab **1416** may be radially offset outwardly from the second inner portion **1426**. Bilateral transverse holes **1430**, **1432** may extend through the release ring **344**. The holes **1430**, **1432** may intersect the second inner portion **1426**.

(276) The instrument **300** may be assembled by coupling the release ring **344**, ratchet **338**, stop screw **316**, and spring plunger **336** to the inserter handle **302**, coupling the lever link **340** and return link **342** to the lever **312**, coupling the sub-assembly of the lever link **340**, the return link **342**, and the lever **312** to the inserter handle **302** (with spring plunger **336**), coupling the shaft assembly **304** and the lock lever **314** to the inserter handle **302**, inserting the lock tube assembly **318** into the shaft assembly **304**, coupling the shuttle **322** to the inserter handle **302** (with stop screw **316** and release ring **344**), inserting the draw bar assembly **328** into the lock tube assembly **318**, and coupling the draw bar assembly **328** to the shuttle **322**. These steps may be performed in any order.

(277) Coupling the release ring **344** to the inserter handle **302** may include inserting the first outer portion **1322** of the inserter handle **302** into the second inner portion **1426** of the hole **1422** of the release ring **344** so that the tab **1416** is in the recess **1334** between the ridges **1336**, **1338**, aligning the holes **1430**, **1432** with the groove **1328**, and inserting pins into the holes **1430**, **1432** and the groove **1328**.

(278) Coupling the ratchet **338** to the inserter handle **302** may include inserting a compression spring (not shown) into the holes **912**, **1354**, inserting the ratchet **338** into the slot **1346** with the tooth **906** facing into the first inner portion **1386** of the hole **1384**, aligning the holes **1356**, **910**, and inserting a pin (not shown) through the holes **1356**, **910**. The spring may bias the ratchet **338** to protrude into the first inner portion **1386**.

(279) Coupling the stop screw **316** to the inserter handle **302** may include inserting the dog point (tip) of the screw into the hole **1358**, engaging the complementary threads, and tightening the screw in the hole **1358**.

(280) Coupling the spring plunger **336** to the inserter handle **302** may include inserting a compression spring (not shown) into the hole **900** of the spring plunger **336** and inserting the proximal end **892** of the spring plunger **336** into the proximal portion of the hole **1372** of the inserter handle **302** so that the second outer portion **898** protrudes distally into the proximal portion **1364** of the slot **1360**.

(281) Coupling the lever link **340** and the return link **342** to the lever **312** may include positioning the distal end **916** of the lever link **340** between the inner arms **884**, **886** of the lever **312**, next to the arm **884**, with the hole **924** aligned with the hole **888** and the tooth **918** facing away from the arm **874**; positioning the distal end **932** of the return link **342** between the inner arms **884**, **886**, next to the arm **886**, with the hole **944** aligned with the hole **888** and the foot **934** facing towards the proximal end **914** of the lever link; and inserting a pin (not shown) through the holes **888**, **924**,

944 to hingedly connect the lever link **340** and the return link **342** to the lever **312**. The pin may only extend through the inner arms **884**, **886**, the lever link **340**, and the return link **342**. A compression spring (not shown) may be inserted in the holes **892**, **926** to bias the lever link **340** away from the lever **312**.

(282) Coupling the sub-assembly of the lever link **340**, the return link **342**, and the lever **312** to the inserter handle **302** (with spring plunger **336**) may include inserting the inner arms **884**, **886** of the lever **312** into the distal portion **1366** of the slot **1360** of the inserter handle **302** so that the inserter handle **302** is between the outer arms **880**, **882** and the lever link **340** and the return link **342** extend into the proximal portion **1364** of the slot **1360**, inserting the second outer portion **898** of the spring plunger **336** into the hole **942** of the return link **342**, and inserting a pin (not shown) into the holes **890**, **1380**. The spring plunger **336** and associated compression spring may bias the lever **312** away from the inserter handle **302**. The ledge **1368** fits in the notch **922** to control the depth to which the tooth **918** extends into the first inner portion **1386** of the hole **1384**.

(283) Coupling the shaft assembly **304** and the lock lever **314** to the inserter handle **302** may include inserting the ring **1398** of the lock lever **314** into the slot **1382** of the inserter handle **302** so that the stem **1400** and cap **1402** extend laterally outside the slot **1382** and the hole **1408** faces the front surface **1330**, inserting the proximal end **978** of the inserter shaft **306** into the third inner portion **1389** of the hole **1384** in the distal end **1320** of the inserter handle **302** and into the hole **1404** of the lock lever **314**, aligning the grooves **986**, **988** with the holes **1390**, **1392**, and inserting pins (not shown) into the holes **1390**, **1392** and grooves **986**, **988**. A compression spring (not shown) may be inserted in the hole **1408** to bear against a front wall of the slot **1382**. The spring may bias the lock lever **314** toward the back surface **1332**.

(284) Inserting the lock tube assembly **318** into the shaft assembly **304** may include inserting the distal end **1246** of the distal lock tube **326** into the hole **992** in the proximal end **978** of the inserter shaft **306**, advancing the lock tube assembly **318** into the shaft assembly **304** until the body **1220** of the lock tube coupling **320** enters the hole **992** and the post **1226** enters the slot **982** or the slot **984**, and twisting the lock tube assembly **318** relative to the shaft assembly **304** to move the post **1226** past the slot bend. When the shaft assembly **304**, lock lever **314**, inserter handle **302**, and lock tube assembly **318** are operatively assembled, the body **1220** of the lock tube coupling **320** may fit in the hole **1404** of the lock lever **314** and the posts **1222**, **1226** of the lock tube coupling **320** may fit in the keyway **1406** of the lock lever **314**.

(285) Coupling the shuttle **322** to the inserter handle **302** (with stop screw **316** and release ring **344**) may include inserting the proximal end **1240** of the proximal lock tube **324** in the fourth inner portion **976** of the hole **968** at the distal end **948** of the shuttle **322**, inserting the key **1428** of the release ring **344** in the groove **958** at the distal end **948** of the shuttle **322**, sliding the shuttle **322** into the first inner portion **1386** of the hole **1384** until the dog point of the stop screw **316** slides into the slot **966**, and twisting the release ring **344** and shuttle **322** relative to the inserter handle **302** to move the dog point past the slot bend.

(286) When the instrument **300** is operatively assembled as described in the preceding steps, the proximal lock tube **324** is in the hole **968** of the shuttle **322**, the tooth **918** of the lever link may engage the grooves **960** of the shuttle **322**, the tooth **906** of the ratchet **338** may engage the grooves **962** of the shuttle **322**, and a portion of the indicia **964** of the shuttle **322** may be exposed through the window **1342** of the inserter handle **302**.

(287) Inserting the draw bar assembly **328** into the lock tube assembly **318** may include inserting the distal end **1288** of the draw bar tip **332** into the inner diameter of the proximal end **1240** of the proximal lock tube **324** and advancing the draw bar assembly **328** into the lock tube assembly **318** until the proximal end **1240** enters the gap **1316**.

(288) Coupling the draw bar assembly **328** to the shuttle **322** may include engaging the internal threads of the second inner portion **972** of the hole **968** of the shuttle **322** with the external threads **1276** of the draw bar handle **330** and tightening the threads until the flange **1272** abuts the proximal

end **946** of the shuttle **322**.

(289) The instrument **300** may be operated by rotating the release ring **344**, pivoting the lock lever **314**, and pivoting the lever **312**.

(290) The release ring **344** may be coupled to the shuttle **322** by the key **1428** in the groove **958** so that the shuttle is free to slide axially relative to the release ring, and the shuttle is rotationally coupled to the release ring so that rotation of the release ring causes rotation of the shuttle. When the release ring **344** is rotated so that the groove **1420** engages the ridge **1338**, the slot **966** of the shuttle **322** may be free to slide axially relative to the stop screw **316** and the grooves **960**, **962** may rotate out of engagement with the teeth **918**, **906**, respectively. When the release ring **344** is rotated so that the groove **1418** engages the ridge **1336**, the slot **966** may capture the stop screw **316** past the slot bend. Thus the release ring **344** and the stop screw **316** cooperate to connect or disconnect the shuttle **322** and the inserter handle **302**.

(291) The lock lever **314** may be coupled to the lock tube coupling **320** by the posts **1222**, **1226** in the keyway **1406** so that the lock tube coupling is free to slide axially relative to the lock lever, and the lock tube coupling is rotationally coupled to the lock lever so that rotation of the lock lever **314** causes rotation of the lock tube coupling. When the lock lever **314** is rotated toward the front surface **1330** of the inserter handle **302**, the posts **1222**, **1226** may be free to slide axially relative to the keyway **1406** and the slot **982** or **984** of the inserter shaft **306**. When the lock lever **314** is rotated toward the back surface **1332**, the slot **982** or **984** may capture the post **1226** past the slot bend. The lock lever **314** may be biased toward the back surface **1332**. The lock lever **314** thus connects or disconnects the lock tube assembly **318** and the shaft assembly **304**.

(292) The lever **312** and associated ratchet mechanism **310** may be coupled to the other parts of the instrument **300** so that actuation of the lever causes proximal axial translation of the shuttle **322** relative to the inserter handle **302**. Pivoting the lever **312** away from the inserter handle **302** may move the lever link **340** distally relative to the shuttle **322**. Pivoting the lever **312** toward the inserter handle **302** may move the lever link **340** proximally. When the tooth **918** is engaged in a groove **960**, pivoting the lever **312** toward the inserter handle **302** also may cause linear translation of the shuttle **322** proximally along the longitudinal instrument axis **866** without rotation of the shuttle **322** about the axis **866**. Thus the lever link **340** may be referred to as a drive link of the ratchet mechanism **310**. The tooth **906** of the ratchet **338** may engage the grooves **962** to prevent the shuttle **322** from moving distally.

(293) Referring to FIGS. **86-91**, another insertion and expansion instrument is shown. Instrument **1500** may be deployed to insert and expand implant **1000**, implant **1100** as shown, or another expandable intervertebral implant. Other tools such as the draw bar assembly **160**, funnel **170**, tamp assembly **180**, and screwdriver assembly **190** disclosed herein may be used, or adapted for use, with instrument **1500** to deploy an implant, deposit graft material within the implant, and secure a locking fastener such as screw **654**.

(294) The instrument **1500** may differ from instruments **100** and **300** in at least the following attributes. The draw bar may be engaged by a slider button/lever that may prevent the draw bar from rotating, and may engage the rotating portion of the inserter handle to enable expansion. The slider button/lever may reduce the potential for incorrect user inputs, i.e., poka-yoke. The draw bar length may be adjustable so that one draw bar may be compatible with multiple implant lengths. The handle may be removable from the tube portion of the instrument **1500** to minimize the mass attached to the implant during bone graft insertion or other steps in the methods of use. Minimizing the instrument mass attached to the implant, and the associated moment arm through the center of gravity of the instrument mass, may minimize unintended motion or misalignment of the implant during steps in the methods of use. The funnel may be used to unthread the inserter shaft (implant retainer shaft) from the implant, for example, after installation of the lockout screw **654**.

(295) The instrument **1500** may include an inserter connector **1502** which may be similar to inserter connector **138** or distal tip part **308**, a connector plate **1504** which may be similar to

impaction cap **146**, an inserter shaft **1506** which may be similar to inserter shaft **136** or **306**, an implant retainer shaft **1508** which may function similar to lock tube assembly **118** or **318**, a connector button **1510**, a thumb wheel **1514**, a wheel axle **1516**, an inserter handle **1518**, a knob strike plate **1520**, a spline insert **1522**, a rotation lock ring **1524**, a cam lever **1526**, a slider button **1528**, a threaded slider **1530**, a top knob part **1532**, a bottom knob part **1534**, and a draw bar assembly **1536** which may be similar to draw bar assembly **160** or **328**. The instrument **1500** may also include various pins, screws and/or bolts, springs, ball detents, and the like. These parts may be basic off-the-shelf components, and they are generally not shown in this disclosure. The instrument **1500** extends from a proximal end **1538** to a distal end **1540** along a longitudinal instrument axis **1542**.

(296) Referring to FIGS. **92** and **93**, the inserter connector **1502** extends between a proximal end **1544** and a distal end **1546**. The proximal end **1544** may be circular in a proximal view. The distal end **1546** may be rectangular in a distal view. The distal end **1546** may have rounded corners in the distal view. The inserter connector **1502** may include, from proximal to distal, first, second, and third outer portions **1548**, **1550**, **1552**. The first outer portion **1548** may have a circular cross section and may extend distally about one fourth to one third of the proximal-distal length of the inserter connector **1502**. The second outer portion **1550** may be a curvaceous transition surface that blends between the first and third outer portions **1548**, **1552**. The third outer portion **1552** may include the rectangular distal end **1546**. An outer distal edge of the rectangle may be chamfered. The inserter connector **1502** may include a central longitudinal hole **1554** that extends through the distal tip part between the proximal end **1544** and the distal end **1546**. The hole **1554** may include, from proximal to distal, first, second, and third inner portions **1556**, **1558**, **1560**. The first inner portion **1556** may have a circular cross section and may extend distally about one fourth to one third of the overall proximal-distal length of the inserter connector **1502**. The second inner portion **1558** may have a circular cross section with a diameter that is less than the diameter of the first inner portion **1556**. The cross sectional shape of the third inner portion **1560** may be the superposition of a circle and a rectangle. The diameter of the circle may be less than the diameter of the first inner portion **1556** and the same as or similar to the diameter of the second inner portion **1558**. The major dimension (length) of the rectangle may be greater than the diameter of the first inner portion **1556**, and may be aligned with the major dimension (length) of the rectangle of the third outer portion **1552**. The minor dimension (width) of the rectangle may be less than the diameter of the circle, and may be less than the diameter of the second inner portion **1558**. The rectangle may have rounded corners. The third inner portion **1560** may have a flat proximal end.

(297) Referring to FIGS. **94** and **95**, the connector plate **1504** extends between a proximal side **1562** and a distal side **1564**. The connector plate **1504** may be circular in a proximal or distal view. The connector plate **1504** may include, from proximal to distal, first and second outer portions **1566**, **1568**. The first outer portion **1566** may taper inwardly from proximal to distal, and may be conical.

(298) The first outer portion **1566** may have a flat distal end. The second outer portion **1568** may be cylindrical, with a diameter that is less than the minor diameter of the first outer portion **1566**. Bilateral holes **1570**, **1572** may extend through the connector plate **1504** between the proximal and distal sides **1562**, **1564**. A central longitudinal hole **1574** may extend through the connector plate **1504** between the proximal and distal sides **1562**, **1564**. The hole may include, from proximal to distal, first, second, third, fourth, and fifth inner portions **1576**, **1578**, **1580**, **1582**, **1584**. Each inner portion may have a circular cross section. The first inner portion **1576** may have a short proximal-distal length and a circumferential chamfer around an inner proximal edge. The second inner portion **1578** may have a diameter that is greater than the diameter of the first inner portion **1576**. The third inner portion **1580** may have a diameter that is the same as or similar to the diameter of the first inner portion **1576**, and less than the diameter of the second inner portion **1578**. There may be a tapered transition between the second and third inner portions **1578**, **1580**. The distal end of

the third inner portion **1580** may be flat and may be proximal to the distal end of the first outer portion **1566**. The fourth inner portion **1582** may have a diameter that is less than the diameter of the first inner portion **1576**. The fifth inner portion **1584** may have a diameter that is less than the diameter of the first inner portion **1576** and greater than the diameter of the fourth inner portion **1582**.

(299) Referring to FIGS. **96** and **97**, the inserter shaft **1506** extends between a proximal end **1586** and a distal end **1588**. The inserter shaft **1506** may be a tubular part with circular outer and inner cross-sectional shapes. The inserter shaft **1506** may include, from proximal to distal, first, second, and third outer portions **1590**, **1592**, **1594**. The first outer portion **1590** may have a short proximal-distal length. The outer diameter of the second outer portion **1592** may be greater than the outer diameter of the first outer portion **1590**. The second outer portion **1592** may occupy most of the overall proximal-distal length of the inserter shaft **1506**. The second outer portion **1592** may include one or more openings **1596**. Four openings **1596** are shown evenly spaced along the length of the second outer portion **1592**. Each opening **1596** may extend through one wall of the second outer portion **1592**, or through both walls as shown. The outer diameter of the third outer portion **1594** may be the same as or similar to the outer diameter of the first outer portion **1590** and less than the outer diameter of the second outer portion **1592**. The inserter shaft **1506** may include a central longitudinal hole **1598** that extends through the inserter shaft between the proximal end **1586** and the distal end **1588**.

(300) The inserter connector **1502**, connector plate **1504**, and inserter shaft **1506** may be coupled together in a shaft assembly **1600** like shaft assembly **116** or **304**. See FIGS. **86** and **87**. The shaft assembly **1600** may also include two pins (not shown) that protrude proximally from the holes **1570**, **1572** in the connector plate **1504**. The shaft assembly **1600** may be assembled by inserting the first outer portion **1590** of the inserter shaft **1506** in the fourth inner portion **1582** of the connector plate **1504** and inserting the third outer portion **1594** in the first inner portion **1556** of the inserter connector **1502**. The holes **1570**, **1572** may be aligned adjacent to the long sides of the rectangle of the third outer portion **1552**. The shaft assembly **1600** may be a weldment. The diameters of the second inner portion **1558** of the inserter connector **1502**, the fourth inner portion **1582** of the connector plate **1504**, and the hole **1598** of the inserter shaft **1506** may be the same or similar, and together, they may form a shaft assembly bore.

(301) Referring to FIGS. **98** and **99**, the implant retainer shaft **1508** extends between a proximal end **1602** and a distal end **1604**. The implant retainer shaft **1508** may include, from proximal to distal, first, second, and third outer portions **1606**, **1608**, **1610**. The first outer portion **1606** may be an elongated torque fitting, such as the hexagonal key shown. The second outer portion **1608** may extend over most of the proximal-distal length of the implant retainer shaft **1508**, and may have a circular cross section with a diameter that is less than the minor diameter of the first outer portion **1606**. The third outer portion **1610** may include external threads. The major diameter of the threads may be the same as or similar to the diameter of the second outer portion **1608**. A central longitudinal hole **1612** may extend through the implant retainer shaft **1508** between the proximal and distal ends **1602**, **1604**.

(302) Referring to FIGS. **100** and **101**, the connector button **1510** extends between a proximal end **1614** and a distal end **1616**. The connector button **1510** may have a flat inner side **1618** and an opposite outer side **1620**. A hole **1622** may extend into the inner side **1618**. A transverse groove **1624** may extend across the outer side **1620** near the proximal end **1614** so as to form a corresponding ledge **1626** along the outer proximal edge of the connector button **1510**. Bilateral thin ears **1628**, **1630** may be located at either end of the ledge **1626**, forming the inner proximal corners of the connector button **1510**. A dimple **1632** may be present on the outer side **1620**. Bilateral slots **1634**, **1636** may extend through the connector button **1510** parallel to the inner side **1618** and through the ledge **1626**. The slots **1634**, **1636** may intersect the groove **1624**. A second connector button **1512** is shown. The connector button **1512** may be a mirror image of the

connector button **1510**.

(303) Referring to FIGS. **102** and **103**, the thumb wheel **1514** extends between a proximal side **1638** and a distal side **1640**. The thumb wheel **1514** may be a generally disc- or ring-shaped part with one or more friction features **1642** arranged around its outer perimeter, such as the ten scallops shown. A central longitudinal hole **1644** may extend through the thumb wheel **1514** between the proximal and distal sides **1638**, **1640**. The hole **1644** may include, from proximal to distal, first and second inner portions **1646**, **1648**. The first inner portion **1646** may have a circular cross section and may extend distally about two fifths to half of the overall proximal-distal length of the thumb wheel **1514**. The second inner portion **1648** may be a torque fitting such as the hexagonal hole shown. The major dimension of the second inner portion **1648** may be less than the diameter of the first inner portion **1646**.

(304) Referring to FIGS. **104** and **105**, the wheel axle **1516** extends between a proximal end **1650** and a distal end **1652**. The wheel axle **1516** may be a funnel shaped part with a wide proximal mouth **1654** and a narrow distal stem **1656** or shaft. The mouth **1654** may taper towards the stem **1656**. A central longitudinal hole **1658** may extend through the wheel axle **1516** between the proximal and distal ends **1650**, **1652**. The hole **1658** may include, from proximal to distal, first, second, and third inner portions **1660**, **1662**, **1664**. The first inner portion **1660** may have a round cross sectional shape and may correspond to the outer shape of the mouth **1654**. The second inner portion **1662** may have a circular cross sectional shape and may extend distally into the stem **1656**. The third inner portion **1664** may have a circular cross section with a diameter that is greater than the diameter of the second inner portion **1662**.

(305) Referring to FIGS. **106-108**, the inserter handle **1518** extends between a proximal end **1668** and a distal end **1670**. The inserter handle **1518** may include, from proximal to distal, first, second, third, and fourth outer portions **1672**, **1674**, **1676**, **1678**. The first outer portion **1672** may have a circular cross section and a short proximal-distal length. The second outer portion **1674** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1672**. The third outer portion **1676** may be generally hourglass shaped or otherwise contoured to be comfortably and ergonomically gripped by a user's hand. The proximal diameter of the third outer portion **1676** may be greater than the diameter of the first outer portion **1672**, so that the second outer portion **1674** forms a circumferential groove between the first and third outer portions **1672**, **1676**. The fourth outer portion **1678** may have a circular cross section with a diameter that is less than the diameter of the second outer portion **1674** and the distal diameter of the third outer portion **1676**. A transverse hole **1680** may extend through the inserter handle **1518** in the third outer portion **1676** along a first direction, which will be referred to as a front-back direction. The hole **1680** may be rectangular or square and may have rounded corners. A pocket **1682** may extend transversely into the inserter handle **1518** along a second direction, which will be referred to as a right-left direction, which is 90 degrees from the first direction. The pocket **1682** may extend proximally into the third outer portion **1674** and distally into the fourth outer portion **1678**. The pocket may be D-shaped, with a flat distal side and a curved proximal side. A second pocket **1684** may extend into the inserter handle **1518** opposite the pocket **1682**, and may be a mirror image of the pocket **1682**. A central longitudinal hole **1686** may extend through the inserter handle **1518** between the proximal and distal ends **1668**, **1670**. The hole **1686** may include, from proximal to distal, first, second, third, and fourth inner portions **1688**, **1690**, **1692**, **1694**. The first inner portion **1688** may have a cross section that includes a circular portion that is concentric with the second, third, and fourth inner portions **1690**, **1692**, **1694**, and an alcove portion **1696** that extends radially away from the circular portion toward a first side of the inserter handle **1518**, which will be referred to as a front side. The distal end of the first inner portion **1688** may be flat. A hole **1698** may extend distally within the alcove portion **1696**, and may be centered within the alcove portion. The hole **1698** may include a counterbore **1700**. The second inner portion **1690** may taper inwardly as it extends distally. The second inner portion **1690** may be conical. The major diameter of the second

inner portion may be less than the diameter of the circular portion of the first inner portion **1688**. The third inner portion **1692** may have a circular cross section and may extend distally to the hole **1680**. The fourth inner portion **1694** may have a circular cross section with a diameter that is the same as or similar to the diameter of the third inner portion **1692**. Bilateral holes **1702**, **1704** may extend proximally into the distal end **1670** of the inserter handle **1518**. The holes **1702**, **1704** may be located toward the front and back. Bilateral holes **1706**, **1708** may extend proximally into the distal end **1670**, and into the proximal wall of the pocket **1682**. Bilateral holes **1710**, **1712** may extend proximally into the distal end **1670**, and into the proximal wall of the pocket **1684**.

(306) Referring to FIGS. **109** and **110**, the knob strike plate **1520** extends between a proximal end **1714** and a distal end **1716**. The knob strike plate **1520** may have flat left and right sides **1718**, **1720** and an hourglass shape in a left or right view. The knob strike plate **1520** may include one or more openings **1722**. Two openings **1722** are shown extending along a front to back direction. Each opening **1722** may extend through one wall of the knob strike plate **1520**, or through both walls as shown. A central longitudinal hole **1724** may extend through the knob strike plate **1520** between the proximal and distal ends **1716**. The hole **1724** may include, from proximal to distal, first, second, and third inner portions **1726**, **1728**, **1730**. The first inner portion **1726** may include a generally circular central portion with bilateral longitudinal grooves **1732**, **1734** extending along the front and back, so that the first inner portion **1726**, taken as a whole, has a noncircular cross section. The first inner portion **1726** may extend distally over most of the proximal-distal length of the knob strike plate **1520**, and may have a flat distal end. The second inner portion **1728** may have a circular cross section with a diameter that is less than the diameter of the circular central portion of the first inner portion **1726**. The second inner portion **1728** may have a short proximal-distal length. Bilateral holes **1736**, **1738** may extend distally through the second inner portion **1728**, and may be aligned with the grooves **1732**, **1734**. The holes **176**, **1738** may be countersunk. The third inner portion **1730** may have a generally rectangular cross section, and may have rounded corners. The major dimension (length) of the rectangle may be greater than the diameter of the circular central portion of the first inner portion **1726**, and may extend along a front to back direction. The third inner portion **1730** may extend through the front side of the knob strike plate **1520**, so that the front side has a distal notch. The minor dimension (width) of the rectangle may be less than the diameter of the circular central portion of the first inner portion **1726** and greater than the diameter of the second inner portion **1728**. Bilateral teeth **1740**, **1742** may protrude toward each other from the left and right walls of the third inner portion **1730** near the front side.

(307) Referring to FIGS. **111** and **112**, the spline insert **1522** extends between a proximal end **1744** and a distal end **1746**. The spline insert **1522** may have a cylindrical body **1748** with a unilateral tab **1750** extending radially outward from the proximal end **1744**. A longitudinal hole **1752** may extend through the tab **1750**. The hole **1752** may be countersunk on the distal side of the tab **1750**. A central longitudinal hole **1754** may include, from proximal to distal, first and second inner portions **1756**, **1758**. The first inner portion **1756** may have a circular cross section and may include a shallow countersink or chamfer around an inner proximal edge. The second inner portion **1758** may include a torque fitting such as the internal spline as shown.

(308) Referring to FIGS. **113** and **114**, the rotation lock ring **1524** extends between a proximal end **1760** and a distal end **1762**. The rotation lock ring **1524** may include, from proximal to distal, first, second, third, and fourth outer portions **1764**, **1766**, **1768**, **1770**. The first outer portion **1764** may have a circular cross section and may include a rounded outer proximal edge. The second outer portion **1766** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1764**. The third outer portion **1768** may have a circular cross section with a diameter that is the same as or similar to the diameter of the first outer portion **1764**. A distal outer edge of the third outer portion may include a circumferential chamfer. The fourth outer portion **1770** may include a torque fitting such as the external spline shown. The fourth outer portion **1770** may be complementary to the second inner portion **1758** of the spline insert **1522**. A central

longitudinal torque fitting **1772** may extend through the rotation lock ring **1524** between the proximal and distal ends **1760**, **1762**. The torque fitting **1772** may be a hexagonal hole as shown. The proximal end **1760** of the rotation lock ring may be concave. The spline insert **1522** and the rotation lock ring **1524** together may be referred to as a rotation locking mechanism.

(309) Referring to FIGS. **115** and **116**, the cam lever **1526** extends between a proximal end **1774** and a distal end **1776**. The cam lever **1526** may include, from proximal to distal, first and second portions **1778**, **1780**. The first portion **1778** may include a hole **1782**, a first notch **1784**, a protrusion **1786**, a second notch **1788**, and a tooth **1790**. The hole **1782** may extend transversely through the cam lever **1526**. The hole **1782** may be near the proximal end **1774**. A proximal/outer bearing surface **1783** may be located next to the hole **1782**; the surface **1783** may be referred to as a first fulcrum of the cam lever **1526**. The first notch **1784** and the protrusion **1786** may extend transversely across an outer side of the cam lever **1526** distal to the hole **1782**. The protrusion **1786** may be directly adjacent to a distal or outer side of the first notch **1784**. The protrusion **1786** may be referred to as a second fulcrum of the cam lever **1526**. The second notch **1788** and the tooth **1790** may extend transversely across an inner side of the cam lever **1526** distal to the hole **1782**. The tooth **1790** may be directly adjacent to a proximal or inner side of the second notch **1788**. The second notch **1788** and the tooth **1790** may be distal to the first notch **1784** and the protrusion **1786**. The second portion **1780** may be a distally extending arm for actuation by a user. The second portion **1780** may be wider than the first portion **1778** so that rounded shoulders **1792**, **1794** are formed at the border between the first and second portions **1778**, **1780**. The shoulders **1792**, **1794** may face proximally and inwardly.

(310) Referring to FIGS. **117-119**, the slider button **1528** extends between a proximal end **1796** and a distal end **1798**, and an outer side **1800** and an inner side **1802**. The outer side **1800** may include an oval domed surface **1804** with a longitudinal central trough **1806**. The trough **1806** may be generally rectangular and may extend through the distal end **1798**. A window **1808** may extend through the slider button **1528** between the outer and inner sides **1800**, **1802**. The window **1808** may be rectangular, with its major dimension (length) extending along a proximal-distal direction. A flange **1810** may encircle the surface **1804** and may serve as a boundary between the outer and inner sides **1800**, **1802**. The flange **1810** may be discontinuous in the vicinity of the window **1808**. The inner side **1802** may include, from proximal to distal, first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, and tenth portions **1812**, **1814**, **1816**, **1818**, **1820**, **1822**, **1824**, **1826**, **1828**, **1830**. The first portion **1812** may be parallel to the flange **1810** at a first distance from the inner side of the flange **1810**. A hole **1832** may extend outwardly into the first portion **1812** near the proximal end **1796**. The second portion **1814** may slope between the first and third portions **1812**, **1816**. The third portion **1816** may be parallel to the flange **1810** at a second distance from the inner side of the flange, wherein the second distance is less than the first distance. The fourth portion **1818** may slope between the third and fifth portions **1816**, **1820**. The fourth portion **1818** may be parallel to the second portion **1814**. The fifth portion **1820** may be parallel to the flange **1810** at the same or similar level with the outer side of the flange. The sixth portion **1822** may slope between the fifth and seventh portions **1820**, **1824**. The seventh portion **1824** may be parallel to the flange **1810** and at the same or similar level with the third portion **1816**. The eighth portion **1826** may curve between the seventh and ninth portions **1824**, **1828**. The ninth portion **1828** may be parallel to the flange **1810** at a third distance from the inner side of the flange **1810**, wherein the third distance is greater than the first distance. The tenth portion **1830** may protrude a fourth distance from the inner side of the flange **1810**, and may extend distally past the flange. The tenth portion **1830** may include a semicircular yoke **1834** that opens away from the flange **1810**.

(311) Referring to FIGS. **119** and **120**, the threaded slider **1530** extends between a proximal end **1836** and a distal end **1838**, and an outer side **1840** and an inner side **1842**. The threaded slider **1530** may be referred to as an actuator. The threaded slider **1530** may include a rectangular body **1844** with a post **1846** extending from the inner side **1842**. The outer side **1840** may ramp

outwardly from proximal to distal. A notch **1848** may extend into the distal outer edge of the body **1844** between bilateral protrusions **1850**, **1852**. A transverse hole **1854** may extend through the threaded slider **1530**, through the protrusions **1850**, **1852** and intersecting the notch **1848**. A shallow hole **1856** or dimple may extend inwardly into the outer side **1840** near the proximal end **1836** in line with the notch **1854**. A longitudinal hole **1858** extends through the threaded slider **1530** between the proximal and distal ends **1836**, **1838**. The hole **1858** may include a first side portion **1860** toward the outer side **1840** and a second side portion **1862** toward the inner side **1842**. The first side portion **1860** may include internal threads around about 180 degrees of arc, i.e., along a half cylinder. The second side portion **1862** may be smooth, D-shaped, and larger than the major diameter of the internal threads of the first portion **1860**.

(312) Referring to FIGS. **122-123**, the top knob part **1532** extends between a proximal end **1864** and a distal end **1866**, and an outer side **1868** and an inner side **1870**. The outer side **1868** may be domed. A window **1872** may extend through the top knob part **1532** between the outer and inner sides **1868**, **1870**. The window **1872** may be oval or rectangular, with its major dimension (length) extending along a proximal-distal direction. The inner side **1868** may be generally concave, and may include, from proximal to distal, first, second, third, fourth, and fifth portions **1874**, **1876**, **1878**, **1880**, **1882**. The first portion **1874** may have a semicircular cross section and a short proximal-distal length. The second portion **1876** may be oval or rectangular, with its major dimension extending along a proximal-distal direction. The second portion **1876** may be aligned with the window **1872**, and may extend over most of the proximal-distal length of the top knob part **1532**. Bilateral longitudinal ledges **1884**, **1886** may extend along the inner edges of the window **1872**. The third portion **1878** may have a semicircular or half-oval cross section and a short proximal-distal length. The first and third portions **1878** may have the same or similar diameters (or radii), with the third portion **1878** offset toward the outer side **1868**. The fourth portion **1880** may have a semicircular cross section and a short proximal-distal length. The diameter of the fourth portion **1880** may be greater than the diameter of the first portion **1874**. The fourth portion **1880** may be concentric with the first portion **1874**. The fifth portion **1882** may have a semicircular cross section and a short proximal-distal length. The diameter of the fifth portion **1882** may be greater than the diameter of the first portion **1874** and less than the diameter of the fourth portion **1880**. The fifth portion **1882** may be concentric with the first portion **1874**. When taken together, the fourth portion **1880** may be referred to as a groove or undercut and the fifth portion **1882** may be referred to as a lip or overhang. Bilateral holes **1888**, **1890** may extend into the inner side **1870** on either side of the third portion **1878**. The holes **1888**, **1890** may include internal threads. The proximal end **1864** may include a protrusion **1892** that is complementary to the third inner portion **1730** of the knob strike plate **1520**. The protrusion **1892** may be generally rectangular and may include bilateral notches **1894**, **1896** to receive the teeth **1740**, **1742**. An internally threaded hole **1898** may extend distally into the protrusion **1892** between the notches **1894**, **1899** and beside the first portion **1874**.

(313) Referring to FIGS. **124** and **125**, the bottom knob part **1534** extends between a proximal end **1900** and a distal end **1902**, and an outer side **1904** and an inner side **1906**. The outer side **1904** may be domed. One or more openings **1908** may extend through the bottom knob part **1534** between the outer and inner sides **1902**, **1904**. Two openings **1908** are shown. Bilateral holes **1910**, **1912** may extend into the outer side **1904** near the distal end **1902**. The holes **1910**, **1912** may include counterbores. A semicircular wall **1914** or shell may protrude from the inner side **1906** near the proximal end **1900** so that a convex side of the wall **1914** faces proximally. The wall **1914** may include a distally extending internally threaded through hole **1916**. A U-shaped or part-oval wall **1918** may protrude from the inner side **1906** near the distal end **1902** so that a convex side of the wall **1918** faces distally. A hole **1920** may extend through the bottom knob part **1534** between the outer and inner sides **1904**, **1906**. The hole **1920** may be located between the distal end of the wall **1914** and the proximal end of the wall **1918**. The hole **1920** may include, from inner to outer, first,

second, and third inner portions **1922**, **1924**, **1926**. The first inner portion **1922** may have a rectangular or square cross section and may extend outwardly over most of the inner-outer thickness of the bottom knob part **1534**. The first inner portion **1922** may have a flat outer end. The second inner portion **1924** may have a circular cross section with a diameter that is less than the minor dimension of the first inner portion **1922**. The second inner portion **1924** may have a flat outer end. The third inner portion **1926** may have a circular cross section with a diameter that is less than the diameter of the second inner portion **1924**. A longitudinal hole **1928** may extend through the bottom knob part **1534** between the proximal and distal ends **1900**, **1902**. The hole **1928** may include, from proximal to distal, first, second, third, and fourth inner portions **1930**, **1932**, **1934**, **1936**. The first inner portion **1930** may have a circular cross section and may extend distally into the proximal region of the wall **1918**. The second inner portion **1932** may have the same or similar diameter (radius) as the first inner portion **1930**, but may be elongated toward the outer side **1904** to form an oval cross section. The second inner portion **1932** may extend through the distal end of the wall **1918**. The third inner portion **1934** may have a semicircular cross section with a diameter that is greater than the diameter of the first inner portion **1930**, and a short proximal-distal length. The fourth inner portion **1936** may have a semicircular cross section with a diameter that is greater than the diameter of the first inner portion **1930** and less than the diameter of the third inner portion **1934**, and a short proximal-distal length. When taken together, the third inner portion **1934** may be referred to as a groove or undercut and the fourth inner portion **1936** may be referred to as a lip or overhang.

(314) The proximal end **1900** may include a protrusion **1938** that is complementary to the third inner portion **1730** of the knob strike plate **1520** when taken together with the protrusion **1892** of the top knob part **1532**. The outer profile of the protrusion **1938** may be rectangular or square. An internally threaded hole **1940** may extend distally into the protrusion **1938** beside the first inner portion **1930**.

(315) The connector buttons **1510**, **1512**, thumb wheel **1514**, wheel axle **1516**, inserter handle **1518**, knob strike plate **1520**, spline insert **1522**, rotation lock ring **1524**, cam lever **1526**, slider button **1528**, threaded slider **1530**, top knob part **1532**, and bottom knob part **1534** may be coupled together in a handle assembly **1942**. See FIG. 87.

(316) The handle assembly **1942** may be assembled by coupling the connector button **1510** to the inserter handle **1518**; coupling the thumb wheel **1514**, wheel axle **1516**, and spline insert **1522** to the inserter handle **1518**; coupling the cam lever **1526**, slider button **1528**, threaded slider **1530**, rotation lock ring **1524**, and the top knob part **1532** together; coupling the top and bottom knob parts **1532**, **1534** to the inserter handle **1518**; and coupling the knob strike plate **1520** to the top and bottom knob parts **1532**, **1534**.

(317) Coupling the connector button **1510** to the inserter handle **1518** may include inserting a compression spring (not shown) in the hole **1622**, orienting the connector button **1510** and inserter handle **1518** with the proximal ends **1614**, **1668** facing the same direction and the inner side **1618** facing into the pocket **1682**, inserting the connector button **1510** into the pocket **1682**, and inserting pins (not shown) into holes **1706**, **1708** and slots **1634**, **1636**. Coupling the connector button **1512** to the inserter handle **1518** may include similar steps with pocket **1684** and holes **1710**, **1712**.

(318) Coupling the thumb wheel **1514**, wheel axle **1516**, and spline insert **1522** to the inserter handle **1518** may include orienting the thumb wheel **1514** and inserter handle **1518** with the proximal ends **1638**, **1668** facing the same direction, inserting the thumb wheel **1514** into the hole **1680**, inserting a compression spring (not shown) into the third inner portion **1664** of the wheel axle **1516**, inserting the distal end **1652** of the wheel axle **1516** through the third inner portion **1692** of the inserter handle **1518** and into the first inner portion **1646** of the thumb wheel **1516**, inserting the distal end **1746** of the spline insert **1522** into the first inner portion **1688** of the inserter handle **1518**, inserting the tab **1750** into the alcove portion **1696**, and inserting a pin (not shown) into holes **1752**, **1698**.

(319) Coupling the cam lever **1526**, slider button **1528**, threaded slider **1530**, rotation lock ring **1524**, and the top knob part **1532** together may include inserting a ball detent (not shown) in the hole **1832** of the slider button **1528**; inserting the first portion **1778** of the cam lever **1526** through the window **1808** of the slider button from the outer side **1800** so that the second portion **1780** extends outwardly, the protrusion **1786** faces proximally, and the tooth **1790** faces distally; inserting the first portion **1778** into the notch **1848** of the threaded slider **1530** between the protrusions **1850**, **1852** so that the outwardly ramped outer side **1840** abuts the fourth portion **1818**; inserting a pin (not shown) into holes **1854**, **1782**; engaging the yoke **1834** with the first and second outer portions **1764**, **1766** of the rotation lock ring **1524** so that the proximal ends **1760**, **1796** face the same direction; and inserting the second portion **1780** and slider button **1528** into the window **1872** of the top knob part **1532** from the inner side **1870** so that the flange **1810** rests against the ledges **1884**, **1886** and the proximal ends **1864**, **1796** face the same direction.

(320) Coupling the top and bottom knob parts **1532**, **1534** to the inserter handle **1518** may include engaging the first outer portion **1672**, the fourth portion **1880**, and the third inner portion **1934**; engaging the second outer portion **1674**, the fifth portion **1882**, and the fourth inner portion **1936** so that the inner sides **1870**, **1904** are facing; and inserting screws (not shown) in the holes **1910**, **1888** and the holes **1912**, **1890**. When this step is performed after the preceding steps, the post **1846** of the threaded slider **1530** is in the hole **1920** of the bottom knob part **1534**.

(321) Coupling the knob strike plate **1520** to the top and bottom knob parts **1532**, **1534** may include abutting the inner sides **1870**, **1904** so that the proximal ends **1864**, **1900** face the same direction, inserting the protrusions **1892**, **1938** into the third inner portion **1730** so that the notches **1894**, **1896** receive the teeth **1740**, **1742**, and inserting screws (not shown) in the holes **1736**, **1898**, **1916** and the holes **1738**, **1940**.

(322) The handle assembly **1942** may be operated by pressing the connector buttons **1510**, **1512**, rotating the thumb wheel **1514**, pivoting the cam lever **1526**, and rotating the knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534**.

(323) Pressing the connector buttons **1510**, **1512** of the handle assembly **1942** may enable the connector plate **1504** of the shaft assembly **1600** to be disconnected from the handle assembly, and may also enable or facilitate connecting the shaft assembly to the handle assembly. The connector buttons **1510**, **1512** are biased outwardly so that the handle assembly will retain the shaft assembly.

(324) Turning the thumb wheel **1514** may rotate the implant retainer shaft **1508** to connect and disconnect the instrument **1500** and implant **1100**. More specifically, turning the thumb wheel **1514** may engage and disengage the external threads of the third outer portion **1610** of the implant retainer shaft **1508** and the internal threads **1155** of the second end body **1152** of the implant **1100**.

(325) Pivoting the cam lever **1526** may engage and disengage the internal threads of the first side portion **1860** of the hole **1858** of the threaded slider **1530** and the externally threaded portion **2036** of the shaft **2032** of the threaded shaft **1954**, discussed below. Pivoting the cam lever **1526** inwardly (distally) may also lock the slider button **1528** in its distal position, due to the interaction of the tooth **1790** with the distal edge of the window **1808**. Pivoting the cam lever **1526** outwardly may provide sufficient mechanical advantage to overcome residual friction and/or potential energy within the instrument **1500** after expanding the implant **1100**, to disengage the threads of the threaded slider **1530** from the threaded shaft **1954**. The mechanical advantage may be due first to the interaction of the first fulcrum, surface **1783**, with the proximal edge of the window **1808**, and second to the interaction of the second fulcrum, protrusion **1786**, with the proximal portion of the trough **1806** next to the proximal edge of the window **1808**.

(326) Rotating the knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534** relative to the inserter handle **1518** may cause linear translation of the draw bar assembly **1536** along the longitudinal instrument axis **1542** without rotation of the draw bar assembly **1536** about the axis **1542**, due to the engaged threads of the threaded slider **1530** and the threaded shaft **1954**, and the sixth outer portion **1974** of the handle **1944** in the torque fitting **1772** of the rotation lock ring **1524**.

(327) Referring to FIGS. **126-129**, the draw bar assembly **1536** may include a handle **1944**, a pull rod **1946**, a set screw **1948**, a button **1950**, a button spring **1952**, a threaded shaft **1954**, a retainer **1956**, and a retainer spring **1958**.

(328) The handle **1944** extends between a proximal end **1960** and a distal end **1962**. The handle **1944** may include, from proximal to distal, first, second, third, fourth, fifth, and sixth outer portions **1964**, **1966**, **1968**, **1970**, **1972**, **1974**. The first outer portion **1964** may have the largest outer diameter of the handle **1944**. Friction features **1976** may be included around the first outer portion **1964**, such as the ten scallops shown. The second outer portion **1966** may have a circular cross section with a diameter that is less than the diameter of the first outer portion **1964**. A transverse hole **1978** may extend through the second outer portion **1966**. The hole **1978** may include, in sequence, first, second, and third inner portions **1980**, **1982**, **1984**. The first inner portion **1980** may be rectangular or square, and may have a flat bottom. The second inner portion **1982** may have a circular cross section with a diameter that is less than the dimension of the first inner portion **1980**. The third inner portion **1984** may also have a circular cross section, with a diameter that is less than the diameter of the second inner portion **1982**. The third outer portion **1968** may have a circular cross section with a diameter less than the diameter of the second outer portion **1966**, and a short proximal-distal length. The fourth outer portion **1970** may include external threads. The fifth outer portion **1972** may be a torque fitting, such as the hex key shown. The dimensions of the torque fitting may be less than the diameters of the first, second, and third outer portions **1964**, **1966**, **1968**. The sixth outer portion **1974** may also be a torque fitting, such as the hex key shown. The dimensions of the sixth outer portion **1974** may be less than the dimensions of the fifth outer portion **1972**. A central longitudinal hole **1986** may extend through the handle **1944** between the proximal and distal ends **1960**, **1962**. The hole **1986** may include, from proximal to distal, first, second, and third inner portions **1988**, **1990**, **1992**. The first inner portion **1988** may have a short proximal-distal length and may include internal threads. The second inner portion **1990** may be smooth with a circular cross section with a diameter that is less than the minor diameter of the threads of the first inner portion **1988**. The third inner portion **1992** may have a circular cross section with a diameter that is less than the diameter of the second inner portion **1990**.

(329) The pull rod **1946** extends between a proximal end **1994** and a distal end **1996**. The pull rod **1946** may include, from proximal to distal, first, second, third, fourth, and fifth portions **1998**, **2000**, **2002**, **2004**, **2006**. The first portion **1998** may have a generally circular cross section and a short proximal-distal length. The second portion **2000** may have the largest cross sectional dimension of the pull rod **1946**. Bilateral longitudinal flats **2008**, **2010** may extend along the first and second portions **1998**, **2000**. The dimension between the flats **2008**, **2010** may be less than the diameter of the first portion **1998**. A unilateral flat **2012** may extend along the second portion **2000** without affecting the first portion **1998**. A transverse notch **2014** may extend into the second portion **2000** through the side with the flat **2012**. Multiple notches may be present, for example a linear array of notches along the side with the flat **2012**. The third portion **2002** may have a circular cross section with a diameter that is less than the diameter of the first portion **1998**, and may be the same as or similar to the dimension between the flats **2008**, **2010**. The third portion **2002** may extend over most of the proximal-distal length of the pull rod **1946**. The fourth portion **2004** may have a circular cross section with a diameter that is less than the diameter of the third portion **2002**. The fifth portion **2006** may include external threads, with the major diameter of the threads the same as or similar to the diameter of the fourth portion **2004**. The pull rod **1946** may include indicia **2016**, such as text, icons, lines, or other markings.

(330) The button **1950** may include a generally rectangular or square body **2018**. A post **2020** may extend from the body **2018**. A hole **2022** may extend through the body, transverse to the post **2020**. The hole **2022** may be rectangular or oval in cross section, with the major dimension of the rectangle or oval extending in line with the post **2020**. A rib **2024** may protrude from an inner wall of the hole adjacent to the post **2020**. The rib **2024** may be oriented transverse to the hole **2022**.

(331) The threaded shaft **1954** extends between a proximal end **2026** and a distal end **2028**. The threaded shaft **1954** may include a proximal head **2030** and a distal shaft **2032** extending from the head. The shaft **2032** may include a proximal smooth portion **2034** and a distal externally threaded portion **2036**. Bilateral longitudinal troughs **2038**, **2040** may extend across the junction between the head **2030** and the smooth portion **2034**. The thread form of the threaded portion **2036** may be a buttress thread for high axial thrust in one direction. The load-bearing thread face may be substantially perpendicular to the thread axis. The load-bearing thread face of the threaded shaft **1954** may face toward the distal end **2028**. A central longitudinal hole **2042** may extend through the threaded shaft **1954** between the proximal and distal ends **2026**, **2028**. The hole **2042** may be a torque fitting such as the hexagonal hole shown.

(332) The retainer **1956** extends between a proximal end **2044** and a distal end **2046**. The retainer **1956** may be a tubular part. Indicia **2048** may be present on the outer surface of the retainer, such as text, icons, graphics, lines, and the like. A central longitudinal hole **2050** may extend through the retainer **1956** between the proximal and distal ends **2044**, **2046**. The hole **2050** may include, from proximal to distal, first, second, third, and fourth inner portions **2052**, **2054**, **2056**, **2058**. The first inner portion **2052** may have a circular cross section. The second inner portion **2054** may include internal threads. The major diameter of the threads may be the same as or similar to the diameter of the first inner portion **2052**. The third inner portion **2056** may have a circular cross section with a diameter that is the same as or similar to the minor diameter of the threads of the second inner portion **2054**. The third inner portion **2056** may have a flat distal end. The fourth inner portion **2058** may have a circular cross section with a diameter that is less than the diameter of the third inner portion **2056**.

(333) The draw bar assembly **1536** may be assembled by inserting the spring **1952** into the second inner portion **1982** of the hole **1978** of the handle **1944**; inserting the button **1950** into the hole **1978** of the handle **1944** so that the post **2020** is received in the third inner portion **1984** and the body **2018** is received in the first inner portion **1980**; depressing the button **1950** while inserting the distal end **1996** of the pull rod **1946** into the hole **1986** in the proximal end **1960** of the handle **1944** and advancing the pull rod **1946** until the notch **2014** receives the rib **2024**; tightening the set screw **1948** in the first inner portion **1988**; inserting the distal end **1962** of the handle **1944** into the hole **2042** in the proximal end **2026** of the threaded shaft **1954** and advancing the handle **1944** until the hole **2042** receives the fifth outer portion **1972**; inserting a retainer spring **1958** in each trough **2038**, **2040**; and inserting the distal end **2028** of the threaded shaft **1954** into the hole **2050** in the proximal end **2044** of the retainer **1956** and advancing the threaded shaft **1954** until the first inner portion **2052** of the retainer **1956** receives the third outer portion **1968** of the handle **1944**, the second inner portion **2054** threads onto the fourth outer portion **1970**, and the third inner portion **2056** receives the head **2030** of the threaded shaft **1954**.

(334) The draw bar assembly **1536** may be operated by depressing the button **1950** while sliding the pull rod **1946** proximally or distally to engage different notches like notch **2014**. Different pull rods may be substituted in the draw bar assembly **1536** as well. Different draw bar lengths and/or configurations may be compatible with different implant lengths (in the collapsed configuration) or configurations (i.e., implants for anterior, posterior, lateral, or transforaminal interbody fusion and the like). The draw bar assembly **1536** may be an improvement in the art because it may be adjustable to provide the draw bar length that corresponds to any given implant length. Even if an incorrect draw bar length is initially selected, the draw bar assembly **1536** may be re-adjusted to the correct draw bar length quickly and easily, and with positive confirmation that the correct length has been chosen. The draw bar assembly **1536** also reduces instrument inventory versus a set of draw bars of different lengths.

(335) The instrument **1500** may be assembled by coupling the implant retainer shaft **1508** to the handle assembly **1942**, coupling the shaft assembly **1600** to the handle assembly **1942**, and coupling the draw bar assembly **1536** to the handle assembly **1942**.

(336) Coupling the implant retainer shaft **1508** to the handle assembly **1942** may include inserting the proximal end **1602** of the implant retainer shaft **1508** into the fourth inner portion **1694** of the hole **1686** at the distal end **1670** of the inserter handle **1518** and advancing the implant retainer shaft **1508** until the first outer portion **1606** is received in the second inner portion **1648** of the hole **1644** of the thumb wheel **1514**.

(337) Coupling the shaft assembly **1600** to the handle assembly **1942** may include inserting the pins protruding proximally from the holes **1570**, **1572** of the connector plate **1504** into the holes **1702**, **1704** of the inserter handle **1518**, inserting the proximal end **1668** of the inserter handle into the hole **1574** in the proximal side **1562** of the connector plate so that the first outer portion **1672** is received in the third inner portion **1580**, the ledge **1626** of the connector button **1510** is received in the second inner portion **1578**, and the first inner portion **1576** is received in the groove **1624** (likewise for connector button **1512**). Optionally, the connector buttons **1510**, **1512** may be depressed during this step.

(338) Coupling the draw bar assembly **1536** to the handle assembly **1942** may include rotating the cam lever **1526** outwardly, inserting the distal end **1996** of the pull rod **1946** into the hole **1724** in the proximal end **1714** of the knob strike plate **1520** and advancing the draw bar assembly **1536** so that the fourth and fifth portions **2004**, **2002** are received in the hole **1612** of the implant retainer shaft **1606**, the sixth outer portion **1974** of the handle **1944** is received in the torque fitting **1772** of the rotation lock ring **1524** and the second inner portion **1662** of the wheel axle **1516**, the threaded portion **2036** of the threaded shaft **1954** is received in the hole **1858** of the threaded slider **1530** and the hole **1928** of the bottom knob part **1534**, and the retainer **1956** is received in the hole **1724** of the knob strike plate **1520**. The draw bar assembly **1536** may slide into the handle assembly **1942** without rotation about the axis **1542**. After the draw bar assembly **1536** is fully inserted into the handle assembly **1942**, the cam lever **1526** may be rotated inwardly to engage the threads of the threaded slider **1530** and the threaded shaft **1954**.

(339) The instrument **1500** may be operated by pressing the connector buttons **1510**, **1512**, rotating the thumb wheel **1514**, pivoting the cam lever **1526**, and rotating the knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534**.

(340) Various additional tools may be used with instrument **1500** to implant and expand an intervertebral implant. The instrument **1500** and selected additional tools may be provided in a kit and together they may be referred to as an instrument set or system. FIGS. **130-139** show additional tools for use with instrument **1500**, including a trial assembly **2060**, a funnel **2062**, and a screwdriver assembly **2064**.

(341) Referring to FIGS. **130-133**, the trial assembly **2060** may include a handle **2066**, a block **2068**, a head **2070**, and a shaft **2072**. A distal end **2074** of the shaft **2072** may be received in a proximal hole **2076** in the head **2070**. The block **2068** may be received in a distal hole **2078** in the handle **2066**. The block **2068** and hole **2078** may have complementary round proximal portions **2080** and complementary rectangular or square distal portions **2082**, **2084**. The block **2068** may be retained in the handle **2066** by a pin (not shown) through holes **2086**, **2088**. A proximal end **2090** of the shaft **2072** may be received in a distal hole **2092** in the block **2068**. The head **2070**, shaft **2072**, and block **2068** may be a weldment. Alternatively, the handle **2066**, block **2068**, head **2070**, and shaft **2072** may be combined in a single part. The trial assembly **2060** may include various markings or other indicia **2094** to convey the dimensions of the trial and the corresponding implant state (for example, expanded or collapsed).

(342) Referring to FIGS. **134** and **135**, the funnel **2062** may include a wide mouth **2096** that tapers inwardly to a narrow stem **2098** or spout. The stem **2098** may include a torque fitting **2100**, which in this example is a hexagonal hole in the distal end of the stem. The torque fitting **2100** may be sized and shaped to receive the first outer portion **1606** at the proximal end **1602** of the implant retainer shaft **1508**. This enables the funnel **2062** to be coupled to the implant retainer shaft **1508** for delivery of graft material, and it enables the funnel **2062** to be used to unthread the implant

retainer shaft **1508**, for example after the lockout screw **654** has been inserted. See FIGS. **157** and **158**.

(343) Referring to FIGS. **27**, **35**, **36**, the tamp assembly **180** may be used with the instrument **1500** and implant **1100** as described above for the instrument **100** and implant **1000**.

(344) Referring to FIGS. **136-139**, the screwdriver assembly **2064** may include a handle **2102** and a shaft **2104**. The handle **2102** may be the same as or similar to the handle **176** of FIGS. **29**, **37**, and **38**. The shaft **2104** may be the same as or similar to the shaft **178**. A proximal end **2118** of the shaft **2104** may be received in a distal end **2120** of the handle **2102**.

(345) Referring to FIGS. **43** and **44**, the lockout screw **654** may be used with the instrument **1500** and implant **1100** as described above for the instrument **100** and implant **1000**.

(346) Referring to FIGS. **140-142**, implant **1100** is expandable along a horizontal or lateral or transverse or medial-lateral axis **104** and a vertical or cephalocaudal axis **106**. Axis **102** is an anterior-posterior axis. Implant **1100** includes a first support member **1140** and a second support member **1130**, the support members connected by a first end body **1150** and a second end body **1152**. First support member **1140** may include an upper support body **1142** and a lower support body **1144**. Second support member **1130** may include an upper support body **1132** and a lower support body **1134**. Internal threads **1155** are included in the second end body **1152** for connection to the implant retainer shaft **1508**. A plurality of link members **1160**, **1162**, **1166**, **1168** pivotably connect each of the first and second end bodies **1150**, **1152** to each of the support members **1130**, **1140**. Lower support body **1144** includes ramped expansion slots **1174**, **1176**, and upper support body **1142** includes ramped expansion slots **1170**, **1172**. A chamber **1122** may exist in the midst of the support bodies **1142**, **1144**, **1132**, **1134**, at least when the implant **1100** is expanded. The implant **1100** is representative of the intervertebral cages described in pending U.S. patent application Ser. No. 15/244,446.

(347) Referring to FIGS. **143-160**, methods are shown for inserting and expanding an expandable intervertebral implant with instrument **1500** and related instruments.

(348) FIGS. **143** and **144** show a step of using the trial assembly **2060** in an intervertebral space to determine a suitable size for the implant **1100**. The trial assembly **2060** may be inserted into the intervertebral space as shown and then rotated 90 degrees about its longitudinal axis to simulate implant vertical expansion.

(349) FIGS. **145** and **146** show a step of coupling the instrument **1500** and implant **1100** together. Note that the cam lever **1526** has been rotated outwardly so that the draw bar assembly **1536** may be inserted into the instrument **1500** and rotated to engage the threads of the fifth portion **2006** of the pull rod **1946** of the draw bar assembly **1536** with internal threads of the first end body **1150** of the implant **1100**.

(350) FIGS. **147** and **148** show a step of locking the instrument **1500** and implant **1100** together. Note that the cam lever **1526** has been rotated inwardly (distally). With the cam lever in this position, the external spline of the fourth outer portion **1770** of the rotation lock ring **1524** is engaged with the internal spline of the second inner portion **1758** of the spline insert **1522** to prevent the draw bar assembly **1536** from rotating about the axis **1542**, and the internal threads of the first side portion **1860** of the hole **1858** of the threaded slider **1530** engage the externally threaded portion **2036** of the threaded shaft **1954** of the draw bar assembly **1536** to prepare the instrument **1500** to be actuated to expand the implant **1100**.

(351) FIGS. **149** and **150** show a step of expanding the implant **1100** laterally. The knob strike plate **1520**, top knob part **1532**, and bottom knob part **1534** may be rotated together in a clockwise direction relative to the inserter handle **1518** to rotate the internal threads of the first side portion **1860** of the hole **1858** of the threaded slider **1530** relative to the externally threaded portion **2036** of the threaded shaft **1954** of the draw bar assembly **1536** to urge the draw bar assembly proximally to expand the implant **1100**.

(352) FIGS. **151** and **152** show a step of expanding the implant **1100** vertically. This step may be

functionally continuous with the step of expanding the implant laterally. Note that implant **1100** features bilateral vertical expansion of the first and second support members **1140**, **1130**.

(353) FIGS. **153** and **154** show a step of releasing the cam lever **1526** after implant expansion is complete. Releasing the cam lever **1526** disengages the external spline of the fourth outer portion **1770** of the rotation lock ring **1524** from the internal spline of the second inner portion **1758** of the spline insert **1522** so that the draw bar assembly **1536** can rotate about the axis **1542**, and disengages the internal threads of the first side portion **1860** of the hole **1858** of the threaded slider **1530** from the externally threaded portion **2036** of the threaded shaft **1954** of the draw bar assembly **1536** by moving the threaded slider **1530** transverse to the axis **1542** so that the smooth second side portion **1862** of the hole **1858** approaches the externally threaded portion **2036**.

(354) FIGS. **155** and **156** show a step of unlocking the instrument **1500** and implant **1100**. This step may be functionally continuous with the preceding step. At this point, the draw bar assembly **1536** may be removed from the instrument **1500**.

(355) FIG. **157** shows a step of removing the handle assembly **1942** from the shaft assembly **1600** and the implant retainer shaft **1508**. The connector buttons **1510**, **1512** are depressed to disconnect the shaft assembly **1600** from the handle assembly **1942**, which also allows the implant retainer shaft **1508** to slide out. Note that the handle assembly **1942** may weigh much more than the shaft assembly **1600** and implant retainer shaft **1508** together, and the handle assembly **1942** is located at the opposite end of the shaft assembly **1600** from the implant **1100**. Thus removing the handle assembly **1942** significantly reduces the force moment on the implant **1100** due to the weight of attached instrument apparatus. Force moment is the force times the moment arm. In the current step, the force is the weight of the handle assembly **1942** due at least to gravity. The moment arm is approximately the length of the shaft assembly **1600** modified with trigonometry to take into account the angle between the shaft assembly **1600** and true vertical, for example for TLIF, the length of the shaft assembly **1600** times the cosine of 30 degrees. The true moment arm is the horizontal distance between the centers of gravity of the implant **1100** and the handle assembly **1942**. Additionally, the removable handle assembly **1942** shortens the length of the screwdriver **2064** and its associated force moment on the implant **1100**.

(356) FIG. **158** shows a step of coupling the funnel **2062** to the implant retainer shaft **1508** (not visible in this figure) and inserting the tamp assembly **180** through the funnel and into the shaft assembly **1600**.

(357) FIG. **159** shows a step of inserting the locking screw **654** using the screw driver assembly **2064** through the funnel **2062** and shaft assembly **1600**. A step of using the funnel **2062** to unscrew the implant retainer shaft **1508** from the implant **1100** is not shown.

(358) FIG. **160** shows a view of the implant **1100** in its final implanted state.

(359) The terms “upper” and “lower”, and “top” and “bottom”, “front” and “rear” are used as relative terms herein for ease of description and understanding. It is understood that in embodiments of the disclosure, upper and lower entities may be reversed, as may top and bottom, front and rear.

(360) Any methods disclosed herein includes one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified.

(361) Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

(362) Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be

interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

(363) Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the technology.

(364) While specific embodiments and applications of the present technology have been illustrated and described, it is to be understood that the technology is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present technology disclosed herein without departing from the spirit and scope of the technology.

Claims

1. An instrument comprising: a handle comprising a distal end comprising a distal thumb wheel and a proximal end comprising a proximal thumb wheel; a first instrument part that is removably connectable to a first implant part of a distal end of an implant via actuation of the proximal thumb wheel; and a second instrument part that is removably connectable to a second implant part of a proximal end of the implant via actuation of the distal thumb wheel; wherein, when the first instrument part is connected to the first implant part, the second instrument part is connected to the second implant part, and the instrument is actuated, the first instrument part and the first implant part translate relative to the second instrument part and the second implant part along a central longitudinal axis of the instrument without rotating about the central longitudinal axis during instrument actuation.

2. The instrument of claim 1, wherein the first instrument part comprises a draw bar that is removably connectable to the first implant part by rotation of the draw bar about the central longitudinal axis; wherein the instrument further comprises an actuator and a rotation lock; wherein instrument actuation comprises movement of the actuator in engagement with the draw bar to translate the draw bar relative to the second instrument part and the second implant part along the central longitudinal axis; wherein when the rotation lock is engaged, the draw bar cannot rotate about the central longitudinal axis, wherein when the rotation lock is disengaged, the draw bar is free to rotate about the central longitudinal axis; wherein the instrument comprises a first state in which the actuator is disengaged from the draw bar, the rotation lock is disengaged, and the draw bar is free to translate along and rotate about the central longitudinal axis; wherein the instrument comprises a second state in which the actuator is engaged with the draw bar, the rotation lock is engaged, and the draw bar is prevented from rotating about the central longitudinal axis.

3. The instrument of claim 2, wherein the rotation lock comprises a rotation lock ring and a spline insert, wherein the rotation lock ring and the draw bar are fixed together for rotation about the central longitudinal axis, wherein the spline insert is fixed in translation along and rotation about the central longitudinal axis, wherein the spline insert is removably connectable to the rotation lock ring; wherein when the rotation lock is engaged, the spline insert is connected to the rotation lock ring so that the rotation lock ring cannot rotate about the central longitudinal axis, wherein when the rotation lock is disengaged, the spline insert is disconnected from the rotation lock ring so that

the rotation lock ring is free to rotate about the central longitudinal axis; wherein the actuator and the rotation lock are coupled together so that the actuator engages the draw bar when the rotation lock is engaged and the actuator disengages from the draw bar when the rotation lock is disengaged.

4. The instrument of claim 3, wherein the draw bar comprises threads; wherein the actuator comprises a threaded slider, wherein the threaded slider comprises threads that engage the threads of the draw bar, wherein instrument actuation comprises rotation of the threaded slider threads in engagement with the draw bar threads to translate the draw bar relative to the second instrument part and the second implant part along the central longitudinal axis.

5. The instrument of claim 1, wherein: the handle further comprises a distal portion and a proximal portion rotatable with respect to the distal portion; and rotation of the proximal portion actuates the instrument.
